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Charles KR, Schults JA, Marsh N, Osborne SR, Rees S, Mihala G, Smith S, Rickard CM.
Clinically indicated removal versus routine removal of peripheral venous catheters.
Cochrane Database of Systematic Reviews 2026, Issue 1. Art. No.: CD007798.
DOI: [10.1002/14651858.CD007798.pub6](https://doi.org/10.1002/14651858.CD007798.pub6).

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[Intervention Review]

Clinically indicated removal versus routine removal of peripheral venous catheters

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Editorial group: Cochrane Central Editorial Service.

Publication status and date: New search for studies and content updated (conclusions changed), published in Issue 1, 2026.

Citation: Charles KR, Schults JA, Marsh N, Osborne SR, Rees S, Mihala G, Smith S, Rickard CM. Clinically indicated removal versus routine removal of peripheral venous catheters. *Cochrane Database of Systematic Reviews* 2026, Issue 1. Art. No.: CD007798. DOI: [10.1002/14651858.CD007798.pub6](https://doi.org/10.1002/14651858.CD007798.pub6).

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ABSTRACT

Rationale

Catheter-related bloodstream infection is a potentially life-threatening complication of peripheral venous catheters (PVCs). Routine PVC removal at set time intervals may prevent bloodstream infection and thrombophlebitis; however, it requires additional replacement PVC insertions and costs. An alternative is clinically indicated removal when the PVC is no longer needed, functional, comfortable or complication-free.

Objectives

To assess the effects of clinically indicated removal versus routine removal of PVCs.

Search methods

We searched the CENTRAL, MEDLINE, Embase and CINAHL databases, and the WHO ICTRP and ClinicalTrials.gov registries to 21 January 2025. We also searched the reference lists of retrieved studies and potentially useful articles.

Eligibility criteria

We included randomised controlled trials (RCTs) comparing clinically indicated with routine PVC removal for patients requiring a PVC. We excluded randomised cross-over trials.

Outcomes

Critical outcomes were catheter-related bloodstream infection, primary bloodstream infection, local infection and mortality.

Important, but not critical, outcomes were thrombophlebitis, infiltration and catheter occlusion.

Outcomes considered of limited importance included cost, pain during infusion, satisfaction and the number of catheters per participant.

Risk of bias

For this update (4th; initial publication 2010), we used the Cochrane risk of bias tool (RoB 1) to assess risk of bias in the studies.

Synthesis methods

Six review authors independently selected studies, extracted data and assessed the risk of bias. Where appropriate, we pooled data using fixed-effect models in the first instance. Effect measures included risk ratios (RR) for categorical outcomes, mean difference (MD) for continuous outcomes and rate ratio (RaR) for incidence rates, each with 95% confidence intervals (CI). We assessed the certainty of evidence using GRADE.

Included studies

All studies in the previous version of the review were included ($n = 9$), and we identified five new studies. The update included 14 RCTs with 11,428 participants. Studies were conducted in Australia, Brazil, China, India, Iran, Taiwan and the United Kingdom. Thirteen studies were conducted in acute hospitals and one in a community setting. Eleven studies assessed adult populations, and three studies included paediatric or neonatal populations. All studies had a high risk of bias in one or more risk of bias domains. The certainty of the evidence was low to moderate, limited by potential detection bias for subjective outcomes, inconsistency across studies and imprecision due to small sample sizes.

Synthesis of results

Clinically indicated removal likely results in little to no difference in the incidence of catheter-related bloodstream infection (RR 0.61, 95% CI 0.08 to 4.68; 10 studies, 10,208 participants; low-certainty evidence) or primary bloodstream infection (RR 0.55, 95% CI 0.22 to 1.38; 3 studies, 6370 participants; low-certainty evidence) compared to routine removal. When compared to routine removal, clinically indicated removal results in little to no difference in local infection risk (RR 2.91, 95% CI 0.46 to 18.39; 9 studies, 8993 participants; low-certainty evidence). There was no clear difference in the incidence of mortality when clinically indicated removal was compared with routine removal (RR 1.01, 95% CI 0.36 to 2.89; 2 studies, 6170 participants; low-certainty evidence). The incidence of all critical outcomes was extremely low for both clinically indicated PVC removal and routine removal.

We are uncertain about the effect of clinically indicated removal of PVCs on thrombophlebitis (RR 1.31, 95% CI 0.94 to 1.82; 14 studies, 10,780 participants; very low-certainty evidence) with little to no difference in thrombophlebitis rates per 1000 catheter-days (RaR 0.98, 95% CI 0.78 to 1.22; 9 studies, 9658 participants; low-certainty evidence). Low-certainty evidence indicated that routine removal has a small effect on reducing infiltration and occlusion per participant, compared with a clinically indicated removal (RR 1.24, 95% CI 1.01 to 1.53; 11 studies, 10,491 participants; RR 1.46, 95% CI 1.06 to 1.99; 12 studies, 10,691 participants respectively). Clinically indicated removal compared with routine removal may result in little to no difference in infiltration rate per 1000 catheter-days (RaR 1.11, 95% CI 0.79 to 1.57; 8 studies, 9458 participants; low-certainty evidence). Compared with routine removal, low-certainty evidence indicated that clinically indicated removal may result in little to no difference in catheter occlusion per 1000 catheter-days (RaR 1.18, 95% CI 0.85 to 1.63; 9 studies, 9658 participants).

Clinically indicated removal of PVCs was associated with a small reduction in device-related costs compared with routine removal (MD -10.88 AUD (Australian dollars) (2024), 95% CI -13.78 to -7.98; 4 studies, 4606 participants). A single study (1319 participants) reported device-related pain and satisfaction results. Clinically indicated removal of PVCs was also associated with a small reduction in the number of catheters required per participant compared with routine removal (MD -0.14, 95% CI -0.25 to -0.03; 8 studies, 8944 participants).

Authors' conclusions

Clinically indicated removal of PVCs makes no difference to the risk of catheter-related bloodstream infection, primary bloodstream infection, local infection or mortality, and likely has little to no effect on thrombophlebitis incidence or rate per 1000 catheter-days, compared to routine removal. The evidence suggests that routine removal may result in a small reduction in the incidence of infiltration and catheter occlusion, but little to no effect on their rates per 1000 catheter-days; however, our confidence in the effect estimate is limited. The true effect may be substantially different from the estimate of the effect. Clinically indicated removal may reduce costs by a small amount compared with routine removal.

Funding

This update of the Cochrane review had no dedicated funding.

Registration

Protocol (2009): <https://doi.org/10.1002/14651858.CD007798>

Original review (2010): <https://doi.org/10.1002/14651858.CD007798.pub2>

Review update (2013): <https://doi.org/10.1002/14651858.CD007798.pub3>

Review update (2015): <https://doi.org/10.1002/14651858.CD007798.pub4>

Review update (2019): <https://doi.org/10.1002/14651858.CD007798.pub5>

PLAIN LANGUAGE SUMMARY

Should we remove peripheral venous catheters ('intravenous drip' or 'cannula') only when they are not needed or have a complication, or should we remove them regularly?

Key messages

- We didn't find any clear difference in bloodstream infection outcomes, or death, between removing PVCs only when needed or regularly, because despite many studies on the topic, serious problems don't happen often.
- The evidence suggests little to no difference in thrombophlebitis (painful vein irritation).
- Infiltration (fluid seeping into the tissue around the PVC) and PVC blockage may be slightly reduced when PVCs are changed regularly.
- Cost is reduced when PVCs are removed for clinical reasons only.
- Future research should focus on testing PVC removal beyond 96 hours, patient preferences for routine or clinically indicated PVC removal, more rigorous studies and use of ultrasound to assess PVC complications, and report outcomes using more sensitive measures (e.g. per 1000 days of PVC use).

What are peripheral intravenous catheters, and what are they used for?

A PVC – also known as an intravenous (IV) drip or IV cannula – is a small hollow tube that is inserted into a small vein close to the surface of the skin. Most hospital patients require a PVC to deliver medication, fluids or nutrition directly into the bloodstream.

Historically, there has been variability in PVC management. PVCs were either removed and replaced every two, three or four days, provided they were still needed, and a complication had not already occurred, or removed/replaced only when a treatment was completed or a complication occurred (clinically indicated).

PVCs often stop working before treatment is completed, and this can happen on any day of treatment. Failure is commonly caused by vein inflammation, catheter blockage, fluids leaking into the surrounding tissues instead of staying inside the vein, or the catheter falling out. Very rarely, serious infections develop. If any of these complications happen, the PVC has to be removed and a new PVC inserted, which can be painful and may cause an interruption/delay in medical treatment.

What did we want to find out?

We wanted to find out if leaving a PVC inserted until medical treatment is complete or PVC complications develop resulted in fewer complications than replacing a PVC every three to four days in people requiring a PVC for at least three days.

What did we do?

We searched medical databases up to 21 January 2025 for randomised controlled trials (a type of study where participants are randomly assigned to one of two or more treatment groups) that compared routine PVC removal with clinically indicated removal.

What did we find?

We found 14 studies involving 11,428 participants comparing clinically indicated versus routine PVC removal. Overall, there was insufficient information to confidently determine whether clinically indicated or routine removal of PVCs was better.

We found that for catheter-related bloodstream infection, bloodstream infection from any cause, local infection and death, there were too few events to determine which method of removal was superior. This means the risk of these problems happening is already extremely low, regardless of clinically indicated or routine removal. Removing the PVC only when clinically indicated might not change vein inflammation, but we can't be sure based on the evidence we have. There's a high chance that the effect could be very different. Changing PVCs regularly might help reduce problems like fluid leaking into the skin or the PVC getting blocked. With clinically indicated removal of peripheral venous catheters, slightly more people may experience complications compared with routine removal: about 37 per 1000 may develop infiltration (range 2 to 83) and about 57 per 1000 may develop catheter blockage (range 7 to 123). These numbers reflect the uncertainty in the evidence. These complications can occur for many people even if there was a planned routine removal. The treatment for these complications is almost always to remove the PVC and replace it if therapy is still required. Removing PVCs only when clinically indicated saves money. The research gives a good idea of this effect, but there's a medium chance that the actual effect could be quite different.

What are the limitations of the evidence?

We considered the overall certainty of the evidence to be low for most results, making the findings somewhat uncertain. We are not confident in the evidence for the studies investigating infection because there were not many infections found during the study. Other

uncertainty is largely due to outcomes, such as thrombophlebitis, catheter blockage and leakage, being assessed by people who were aware of the participant's assigned group, which may or may not affect their decision about whether a problem is present or absent.

How up-to-date is the evidence?

This is the fourth update of a review first published in 2010. The evidence is current to 21 January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Clinically indicated removal of peripheral venous catheters (PVCs) compared to routine removal of PVCs for patients requiring a PVC

Clinically indicated removal of peripheral venous catheters (PVCs) compared to routine removal of PVCs for patients requiring a PVC

Patient or population: patients requiring a PVC

Setting: hospital or community setting

Intervention: clinically indicated removal of peripheral venous catheters (PVCs)

Comparison: routine removal of PVCs

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with routine removal of PVCs	Risk with clinically indicated removal of peripheral venous catheters (PVCs)				
Catheter-related bloodstream infection (CRBSI) assessed with: a positive blood culture from a peripheral vein; clinical signs of infection; no other apparent source for the bloodstream infection except the IV catheter; and colonised IV catheter tip culture with the same organism as identified in the blood	0 per 1000	0 per 1000 (2 to 0)	RR 0.61 (0.08 to 4.68)	10208 (10 RCTs)	⊕⊕⊕⊖ Low ^a	Clinically indicated removal of peripheral venous catheters may result in little to no difference in catheter-related bloodstream infection.
Primary bloodstream infection assessed with: a laboratory-confirmed bloodstream infection drawn from a peripheral vein while an IV catheter is in situ or for 48 hours after removal, with no other identified source of the infection	4 per 1000	2 per 1000 (6 to 1)	RR 0.55 (0.22 to 1.38)	6370 (3 RCTs)	⊕⊕⊕⊖ Low ^a	Clinically indicated removal of peripheral venous catheters may result in little to no difference in primary bloodstream infection.
Local infection assessed with: any definition identified by the trial author	0 per 1000	1 per 1000 (4 to 0)	RR 2.91 (0.46 to 18.39)	8993 (9 RCTs)	⊕⊕⊕⊖ Low ^a	Clinically indicated removal of peripheral venous catheters may result in little to no difference in local infection.

Mortality assessed with: all-cause	2 per 1000	2 per 1000 (7 to 1)	RR 1.01 (0.36 to 2.89)	6170 (2 RCTs)	⊕⊕⊕⊕ Low ^a	Clinically indicated removal of peripheral venous catheters may result in little to no difference in mortality.
Thrombophlebitis - total assessed with: any definition identified by the trial author	87 per 1000	114 per 1000 (158 to 82)	RR 1.31 (0.94 to 1.82)	10780 (14 RCTs)	⊕⊕⊕⊕ Very low ^{b,c,d}	Clinically indicated removal of peripheral venous catheters may have little to no effect on thrombophlebitis but the evidence is very uncertain.
Infiltration assessed with: permeation of IV fluid into the interstitial compartment, causing swelling of, or damage to, local tissue around the site of the catheter	156 per 1000	194 per 1000 (239 to 158)	RR 1.24 (1.01 to 1.53)	10491 (11 RCTs)	⊕⊕⊕⊕ Low ^e	Clinically indicated removal of peripheral venous catheters may result in a slight increase in infiltration.
Catheter occlusion assessed with: the inability to infuse fluids	124 per 1000	182 per 1000 (247 to 132)	RR 1.46 (1.06 to 1.99)	10691 (12 RCTs)	⊕⊕⊕⊕ Low ^e	Clinically indicated removal of peripheral venous catheters may result in an increase in catheter occlusion.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_442956001502329452.

^a Downgraded two levels for very serious imprecision (small number of events and wide confidence intervals).

^b Downgraded two levels due to the overall risk of bias, as three of the 14 studies were assessed to be at high risk across three or more bias domains (Barker 2004; Lu 2021; Nishanth 2009). Barker 2004 (blinding of participants and personnel, blinding of outcome assessor, incomplete outcome data and other bias), Lu 2021 (blinding of outcome assessor, incomplete outcome data and selective reporting) and Nishanth 2009 (uncertain allocation concealment, blinding of outcome assessor and other bias).

^c Downgraded one level due to serious inconsistency (substantial clinical heterogeneity I² = 64%, P = 0.0007).

^d Downgraded one level due to imprecision (95% CI crosses both the null effect (1.0) and multiple thresholds of clinical significance).

^e Downgraded one level due to serious risk of bias (no blind outcome assessment in any of the trials) and one level due to serious inconsistency (substantial heterogeneity).

BACKGROUND

Description of the condition

Peripheral venous catheters (PVC) are the most common vascular access device with a 59% point prevalence in hospitalised patients worldwide [1]. PVCs are indicated for peripherally compatible infusate therapy of up to five days, or up to 14 days if inserted via ultrasound guidance [2]. Many patients are difficult to cannulate due to extremes of age (young or old), obesity, or vein damage from prior treatment and multiple insertion attempts [3, 4]. Once inserted, PVCs can cause vein inflammation due to vein trauma presenting as thrombophlebitis or occlusion [5]. Further, PVCs may become dislodged either into the surrounding tissue (infiltration) or out of the body (dislodgement). As PVCs potentially provide direct entry of microorganisms into the body, they can develop local or catheter-related bloodstream infections (CRBSI) (less than 0.1% [6]). Though infrequent, such infections are serious and can be life-threatening. Overall, one-third of PVCs develop complications causing failure before therapy completion, causing many additional PVC insertions, pain, treatment delays and a heightened risk of complications in subsequent PVCs [6]. This high failure incidence, combined with the large numbers of PVCs inserted, significantly burdens the patient experience, healthcare costs and clinician workloads.

Description of the intervention and how it might work

The underlying mechanisms of PVC complications are (i) vein irritation and damage from the insertion procedure, or subsequent presence of the foreign body and infusates, (ii) microorganism entry into the tissue or blood through the skin wound or the PVC internal lumen, and (iii) movement of the PVC out of the vein (into the tissue or complete dislodgement) due to pressure on the external PVC and inadequate securement to the skin. While more days of PVC use naturally expose the patient to a higher risk of complications, it has been unclear whether removing PVCs at set time periods and inserting replacement PVCs reduces complications for the patient overall, rather than removing and replacing PVCs based on clinical assessment.

Clinically indicated removal requires hospital staff to regularly assess and make decisions about the appropriateness of the PVC and its removal. Removal based on clinical indication includes therapy completion (or switch to oral treatment), patient pain (thrombophlebitis), PVC malfunction or suspected infection. Appropriate PVC removal decisions could reduce complications, unnecessary insertions, clinician workload and resources [7, 8].

Routine removal might prevent infection by removing PVCs that have become colonised, including those with biofilm. However, this triggers additional insertion procedures, which may themselves increase infection risk due to suboptimal skin disinfection and non-aseptic technique. Although routine removal at a set time point does not preclude earlier removal for clinical reasons, in practice, clinical staff often fail to remove PVCs that are not needed or have complications, with a large global point prevalence study finding that 14% were idle (not needed) and 10% had insertion site complications yet had not been removed [9]. Delayed/failed removals can result from lack of clinician confidence that they can insert a new PVC, particularly in difficult patients [10]. Routine removal might encourage a misperception that there is a 'safe' time window for PVC use, making hospital staff less concerned

about infective symptoms that occur before the routine removal time. Another challenge with compliance with the routine removal approach is that up to 50% of PVC insertions do not have the insertion date and time documented in medical records, and many do not have insertion or removal recorded at all; in these cases, calculating dwell time is impossible [9].

Vessel health and preservation during intravenous therapy are crucial considerations. Yet, the additional PVC insertions required to achieve routine removal policies can lead to venous depletion, making future access more difficult and increasing the risk of complications [11]. In practice, hospitals with routine removal policies routinely exempt patients with poor vein quality [12, 13]. Clinically indicated removal aims to safely preserve vein integrity [14], and enhance patient comfort and satisfaction [7].

Why it is important to do this review

Clinically indicated PVC removal is now recommended as first-line management in most international guidelines and standards, aligning PVC removal with other vascular access devices (e.g. central venous catheters) [15, 16, 17]. Routine removal remains practised in many hospitals due to concerns about whether clinically indicated removal will lead to increased infection risk and forgotten or unnecessarily long-dwell PVCs. Due to its very low incidence, PVC-associated bloodstream infection is rarely the primary endpoint of randomised controlled trials (RCTs), leading to imprecise estimates. A pre-post study of a change from routine to clinically indicated PVC removal policy at two hospitals in the USA (N = 1161 patients) included a comprehensive implementation strategy and found a 19% reduction (0.015 to 0.012 per 100 patient days) in PVC-associated bloodstream infections [18]. However, a similar policy change by a large Swiss hospital (N = 164,331 patients) reported increased PVC bloodstream infections from 0.13 to 0.9 cases per 10,000 catheter-days [19]. Although the actual risk per patient in both studies remained low, the difference in the effect direction may reflect the different approaches to implementation strategies, with success requiring a multi-modal implementation strategy including formal staff education about assessment and appropriate criteria for removal [13, 20, 21]. A case series of 99 patients with PVC-BSI from Japan did not find dwell time to be an independent significant risk factor for infection [22].

Thrombophlebitis, infiltration and occlusion are frequent PVC complications with average incidence rates of 39, 33 and 27 per 1000 catheter-days, respectively [5]. All-cause PVC catheter failure may not increase exponentially with dwell time. A prospective observational study in China (N = 1477) reported a bimodal pattern of PVC failure incidence with an increased risk of 1% per hour for the first 38 hours of dwell (incident rate ratio (IRR) 1.01, 95% confidence interval (CI) 1.00 to 1.02), followed by a significant decrease in failure from 39 to 149 hours dwell (IRR 0.994, 95% CI 0.992 to 0.996) [23]. A secondary analysis of multiple prospective studies found that thrombophlebitis occurred at a relatively constant rate over time rather than peaking at a specific point [24]. In contrast, infiltration and occlusion tended to occur earlier in the dwell period. A comprehensive study in intensive care unit (ICU) patients (N = 2741) reported the median time (hours) to phlebitis per patient as 27.1 (interquartile range (IQR) 13.2 to 50.7), suggesting that the majority of PVCs develop phlebitis early in dwell [25]. Guembe et al. also reported no increase in thrombophlebitis with longer dwell time in 70 patients [26].

This is the fourth update of the Cochrane review evaluating the effectiveness of clinically indicated removal for PVCs, initially published in 2010 and previously updated in 2013, 2015 and 2019 [27, 28, 29, 30]. The previous update included nine trials and, since then, an additional five trials have been published.

OBJECTIVES

To assess the effects of clinically indicated removal versus routine removal of PVCs.

METHODS

Since the publication of the original protocol in 2009 [31], we have updated the review methods (see [Supplementary material 8](#)). For this update (2025):

- We approached authors who included patients receiving parenteral nutrition in their cohort for results excluding patients receiving parenteral nutrition (Chin 2018 [32, 33]; Li 2021 [34]).
- We assessed the impact of publication bias with funnel plots for outcomes with ≥ 10 studies.
- We added 'infiltration per catheter-days' and 'occlusion per catheter-days' to the meta-analysis.
- Subgroup analysis was refined to PICO criteria – population (age, high- versus low- to middle-income countries (HIC versus LMIC)), and intervention timing (48-hour routine removal versus 72 hours versus 72 to 96 hours versus 96 hours).
- We changed the 'critical' and 'important, but not critical' outcomes to reflect current endpoint priorities for PVCs [35, 36, 37].
- The summary of findings table now reflects the 'critical' and 'important, but not critical' outcomes.
- We added RCTs with an identified high risk of bias within the sensitivity analyses.
- We removed the outcome 'catheter re-sites per patient' and replaced it with 'number of catheters per participant'.
- We performed a post hoc sensitivity analysis, excluding studies that reported outcomes for only a single catheter per patient, rather than accounting for the entire catheter requirement.

We followed the Methodological Expectations for Cochrane Intervention Reviews (MECIR) when conducting the review [38], and PRISMA 2020 for the reporting [39].

Criteria for considering studies for this review

Types of studies

We included all randomised controlled trials (RCTs) comparing routine removal of PVC with removal when clinically indicated. We excluded cross-over trials to maintain consistency and avoid potential confounding from carryover effects. Although such designs may increase power or reduce contamination, and are permitted in the *Cochrane Handbook for Systematic Reviews of Interventions* [40], their inclusion would have introduced heterogeneity in design and analysis.

Types of participants

We included any patient requiring a PVC to be in situ for at least three days for the administration of intermittent or continuous therapy (this may include patients in hospitals, nursing homes or

in community settings). We excluded patients receiving parenteral nutrition.

Types of interventions

We included short PVCs made of any material and covered by any dressing (for example, gauze or polyurethane). We included clinically indicated removal versus any time duration for routine removal (e.g. 48 or 72 hours). We excluded steel catheters, midline catheters and long peripheral catheters.

Outcome measures

The critical and important outcomes in this review include PVC endpoints considered adverse events for PVCs.

Critical outcomes

- Catheter-related bloodstream infection (CRBSI, defined as a positive blood culture from a peripheral vein; clinical signs of infection; no other apparent source for the bloodstream infection except the intravenous (IV) catheter; and colonised IV catheter tip culture with the same organism as identified in the blood)
- Primary bloodstream infection (defined as a laboratory-confirmed bloodstream infection drawn from a peripheral vein while an IV catheter is in situ or for 48 hours after removal, with no other identified source of the infection)
- Local infection (any definition identified by the trial author)
- Mortality (all-cause)

Important outcomes

Important, but not critical

- Thrombophlebitis/phlebitis (any definition identified by the trial author)
- Infiltration (defined as permeation of IV fluid into the interstitial compartment, causing swelling of, or damage to, local tissue around the site of the catheter)
- Catheter occlusion (the inability to infuse fluids)

Limited importance

- Cost (materials and labour for PVC insertion or treating complications)
- Pain during infusion (measured by a validated pain assessment scale)
- Satisfaction (measured by any validated satisfaction scale)
- Number of catheters per participant (followed for the total duration of treatment)

Search methods for identification of studies

There was no restriction on language. When non-English language studies were found, we sought initial translation of abstracts for the application of the inclusion and exclusion criteria. Where necessary, we also translated the methods, results and discussion sections for inclusion in the review.

Electronic searches

The University of Queensland and Cochrane Vascular Information Specialists systematically searched the following databases for RCTs and controlled clinical trials without language, publication

year or publication status restrictions. Two University of Queensland Health Information Specialists (independent of one another) ran an updated systematic search on 21 January 2025.

- Cochrane Vascular Specialised Register via the Cochrane Register of Studies (CRS-Web; 2025, Issue 1)
- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 1) via the Cochrane Register of Studies Online (CRSO)
- MEDLINE (Ovid MEDLINE Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE Daily and Ovid MEDLINE (from inception to 21 January 2025)
- Embase Ovid (from inception to 21 January 2025)
- CINAHL Ebsco (from inception to 21 January 2025)

We developed search strategies for other databases from the search strategy designed for MEDLINE. Where appropriate, these were combined with adaptations of the highly sensitive search strategy designed by Cochrane for identifying randomised controlled trials and controlled clinical trials (as described in the *Cochrane Handbook for Systematic Reviews of Interventions* Chapter 4 [41]). Search strategies for major databases are provided in [Supplementary material 1](#).

Searching other resources

We contacted researchers in order to obtain any unpublished data. We searched the reference lists of retrieved studies and potentially useful articles. We also searched the following clinical trial registries:

- ClinicalTrials.gov (clinicaltrials.gov/) (21 January 2025), using the terms peripheral and intravenous and catheter and routine;
- World Health Organization International Clinical Trials Registry Platform (ICTRP; apps.who.int/trialsearch/) (21 January 2025) using the terms peripheral and intravenous and catheter.

Prior to final analysis, we checked all included and eligible studies in PubMed and Retraction Watch to identify any post-publication amendments, including expressions of concern, errata, corrigenda or retractions.

Data collection and analysis

Selection of studies

We used Cochrane's Screen4Me workflow to help assess the search results. Screen4Me comprises three components: known assessments – a service that matches records in the search results to records that have already been screened in Cochrane Crowd and been labelled as an RCT or as not an RCT; the RCT classifier – a machine learning model that distinguishes RCTs from non-RCTs, and if appropriate, Cochrane Crowd – Cochrane's citizen science platform where the Crowd help to identify and describe health evidence.

Pairs of review authors independently reviewed titles and abstracts identified through the search process (KC, JAS, NM, SO, SR, CMR). We retrieved full reports of all potentially relevant trials for further assessment of eligibility based on the inclusion criteria. If we excluded any full texts, we recorded the reasons. We completed a PRISMA flow chart [39] ([Figure 1](#)), to summarise the selection process. Authors CR and NM contributed to the overall review process, including screening and data extraction for studies in which they had no authorship or investigator involvement. For studies in which either was an investigator, all stages of assessment, including study selection, data extraction and risk of bias evaluation, were completed independently by other review authors. We settled differences of opinion by consensus or referral to a third review author. There was no blinding of authorship.

Figure 1. Study flow diagram

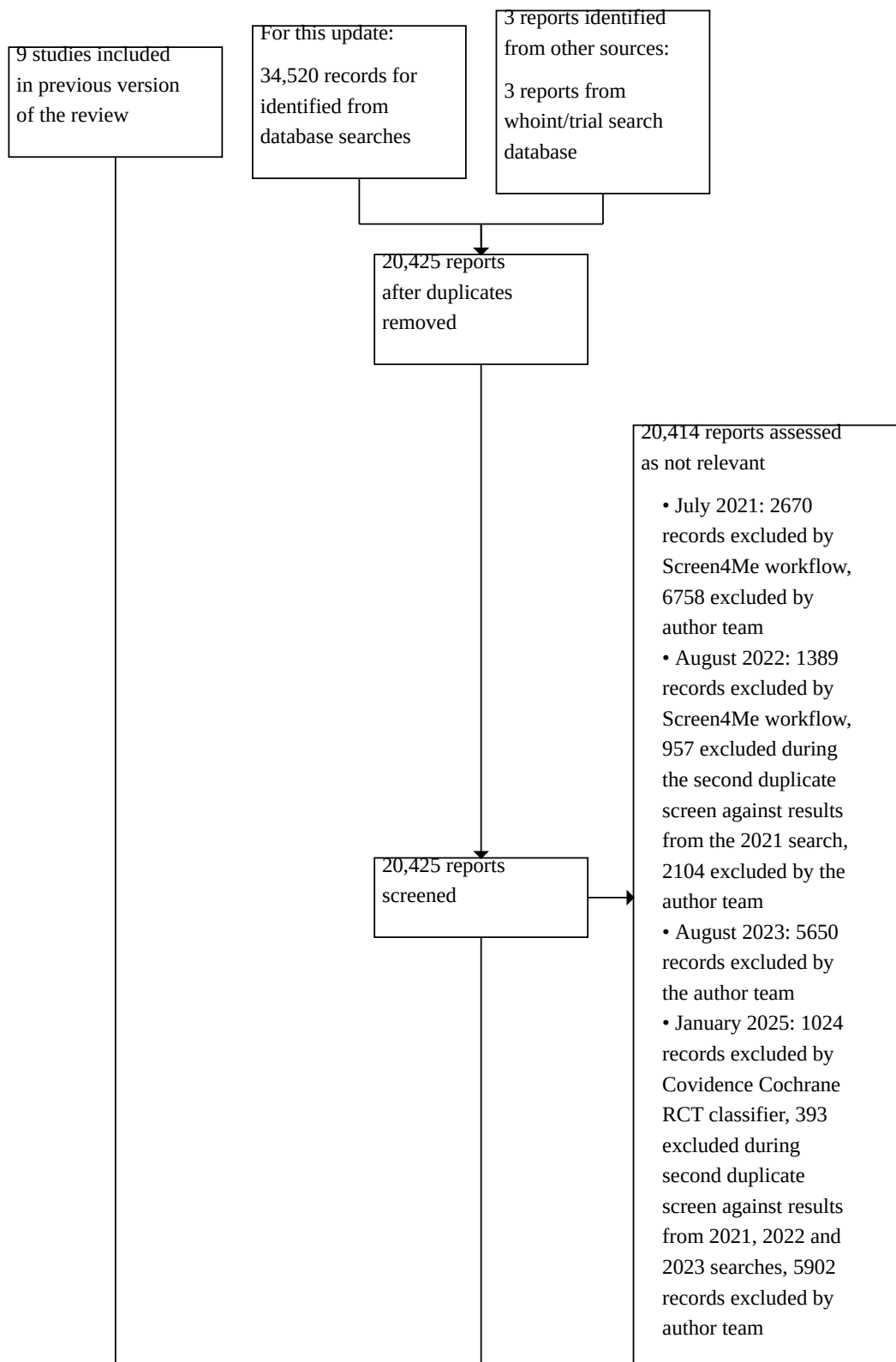
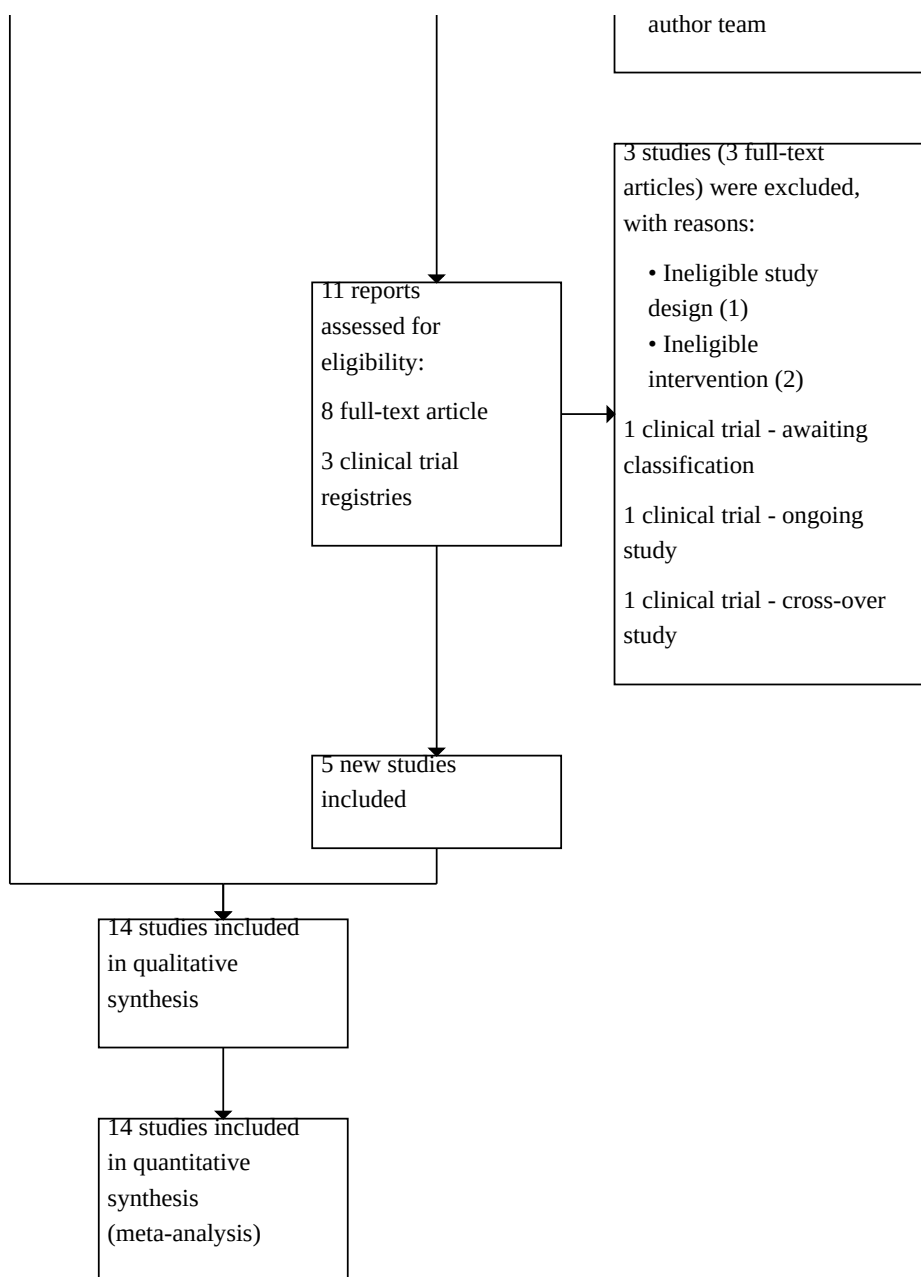


Figure 1. (Continued)



Data extraction and management

Following Cochrane Vascular recommendations, pairs of review authors (KC, JAS, NM, SO, SR, CMR) independently extracted data to a pre-tested data extraction form from prior updates. We resolved disagreements by discussion and, where necessary, by involving a third review author. We contacted published and unpublished trial authors for additional information (KRC and CR). The first review author (KRC) entered the data into Review Manager 5, with a second review author checking data entry accuracy [42]. Authors CMR and NM were investigators in some of the included trials. The assessment and data extraction were allocated to review

authors who were not investigators to eliminate potential conflicts of interest.

Risk of bias assessment in included studies

Pairs of review authors (KC, JAS, NM, SO, SR, CMR) independently assessed the included studies using the Cochrane tool to assess the risk of bias for the following items [43].

- Random sequence generation
- Allocation concealment
- Blinding of participants and personnel
- Blinding of outcome assessment

- Incomplete outcome data
- Selective outcome reporting

We graded each study as high, low or unclear for sources of bias according to the recommendations outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* [43]. We contacted the investigators of the included trials to resolve any ambiguities.

Measures of treatment effect

Effect measures for categorical outcomes included the risk ratio (RR) with its 95% confidence interval (CI). For continuous outcomes, the effect measure was the mean difference (MD) with its 95% CI. For incidence rates, the effect measure was the rate ratio with its corresponding 95% CI. We analysed data using Cochrane statistical software, Review Manager [42].

Unit of analysis issues

A unit of analysis challenge is where trials randomise per patient, but the report includes outcomes for only one PVC per patient since the intervention (removal) reduces the time at risk of complications per PVC. This would bias outcomes in favour of the routine removal group. A typical course of IV therapy is for a week, over which time one or many PVCs may be used, dependent on when they are removed either due to the intervention or due to complications. Ideally, a trial would randomise at the patient level and report outcomes at the patient level, including all PVCs used for the course of treatment. To address this issue, we undertook a sensitivity analysis, removing studies where it was clear that one study group had reported on multiple PVCs and the other had not, or where it was unclear whether this was the approach.

Even if all PVCs are studied per patient, because there is a time-to-event aspect to PVC complications (which are used for differing periods, and may have complications early or late in dwell), then the effect of a removal policy needs to consider time at risk. Thus, appropriate comparisons need to be made that adjust for time at risk; for example, survival analyses or rates per 1000 days. We dealt with this issue by including only studies where data were available per participant rather than per catheter. Where data were not originally analysed in this format, we contacted the investigators (for example, Van Donk 2009 [44]) to obtain these data. Further, we included analyses of thrombophlebitis, infiltration and catheter occlusion per 1000 catheter-days, where this information was available, to ensure that the crude comparisons of outcomes per participant did not lead to erroneous conclusions.

Another challenge is randomising studies at the cluster level or randomising the participant but reporting the number of outcomes per catheter (one patient can have many PVCs). If a cluster trial had been correctly analysed (per patient, not per PVC), we planned to include effect estimates in any meta-analysis using the generic inverse methods in Review Manager [42]. Where a cluster-randomised trial had been incorrectly analysed, we planned to record the unit of analysis issue in our risk of bias assessment. If possible, we planned to approximate the correct analyses based on guidance from the *Cochrane Handbook for Systematic Reviews of Interventions* [40]. That is, to calculate the 'design effect' by $1 + (M-1)$ intraclass correlation coefficient (ICC) where M is the average size of each cluster. We did identify one cluster-randomised trial, the results of which did not appear to have been adjusted (Xu 2017 [45]).

For this trial, we could not identify an ICC, so we used an estimate of 0.02 and calculated the average cluster size to be 59.9.

Dealing with missing data

If any outcome data remained missing despite our attempts to obtain complete outcome data from trial authors, we assessed the risk of bias of missing data and decided whether the missing data were at 'low' or 'high' risk of bias according to our risk of bias criteria [43]. If we considered data missing at random, we analysed the available information. If standard deviations were missing, we planned to impute them from other studies or, where possible, compute them from standard errors using the formula $SD = SE \times \sqrt{N}$ where these were available [46]. For those studies that measured but did not report results as means and standard deviations, we utilised the method of calculating continuous data described by Wan, Wang, Liu and Tong [47]. We performed sensitivity analyses to assess the impact of changing assumptions on the final outcome [46].

Reporting bias assessment

We assessed reporting bias using the guidelines in the *Cochrane Handbook for Systematic Reviews of Interventions* [48]. We pooled outcomes with greater than 10 studies to explore possible small study and publication biases. We assessed reporting bias in each study as part of our risk of bias evaluation.

Synthesis methods

Where clinical heterogeneity, i.e. in populations, interventions and outcomes, was sufficiently low across trials, we pooled results of comparable trials using a fixed-effect model [49]. We reported the pooled estimate together with its 95% CI. We fitted a random-effects model in the analyses with more than 10 studies, or if substantial clinical heterogeneity was present. For the analysis of rates per 1000 catheter-days, the data type was set to 'contrast level', the incidence rate ratio (and 95% CI) was calculated for each study using the `iri` command in Stata (StataCorp. 2023. *Stata Statistical Software: Release 18*. College Station, TX: StataCorp LLC.), then the natural logarithm of the rate ratio and the standard error were entered using RevMan's inbuilt calculator.

For studies reporting the outcome of cost, we intended to convert any foreign currencies to Australian Dollars (AUD) as most previous studies reported cost differences from Australia. For this review update, we adjusted all costs for inflation, current as of available data from the Reserve Bank of Australia, 2024 [50].

When clinical heterogeneity between studies was minimal, but statistical heterogeneity was high ($I^2 > 75\%$), we explored potential sources of the heterogeneity (by examining potentially influential factors, for example, routine removal time interval, care setting or participant characteristics) and noted that the confidence intervals ignored the extent of heterogeneity in interpreting the results. Where statistical synthesis of data from more than one study was not possible or considered inappropriate, we conducted a narrative review of eligible studies [51].

Investigation of heterogeneity and subgroup analysis

We planned a priori subgroup analysis of the critical and the important, but not critical, outcomes to identify potential effect modifiers, regardless of the presence or absence of heterogeneity.

- Population
 - Age group (children < 18 years old versus adults ≥ 18 years old)
 - High- versus low- to middle-income countries (HIC versus LMIC)
- Intervention
 - 48 hours versus 72 hours versus 72 to 96 hours versus 96 hours routine removal interval
 - Continuous versus intermittent infusions

We reported subgroup results against Richardson, Garner and Donegan's recommendations for the interpretation of subgroup analyses in systematic reviews [52].

Equity-related assessment

Acknowledging that context can impact how an intervention is implemented, we collected data on the trial locations (using the country as an indicator of resource level, HIC versus LMIC) and evaluated how contextual factors might affect the generalisability and relevance of the findings when interpreting the evidence.

Sensitivity analysis

We planned to perform sensitivity analyses to explore the effect of the following criteria on the critical and important, but not critical, outcomes:

- Methodological quality (excluding RCTs with higher risk of bias)
- Concealment of allocation (excluding studies with inadequate concealment or not reported)
- Size of studies (excluding studies with < 100 participants)
- Statement of withdrawals and losses to follow-up in each group (excluding studies that did not account for withdrawals and loss to follow-up)
- Unpublished studies (excluding studies for which results were obtained directly from investigators or registries but not peer-reviewed)

We performed post hoc sensitivity analysis, excluding studies that reported outcomes for only a single catheter per patient rather than accounting for the entire catheter requirement.

Certainty of the evidence assessment

To assess the overall body of evidence, we developed a summary of findings table for the following outcomes.

- CRBSI
- Primary bloodstream infection
- Local infection
- Mortality
- Thrombophlebitis
- Infiltration
- Occlusion

We assessed the certainty of the body of evidence using the five GRADE domains: 1) limitations in design and implementation; 2) indirectness of evidence or generalisability of findings; 3) inconsistency of results; 4) imprecision of results where confidence intervals were wide; and 5) other potential biases, for example, publication bias or high manufacturer involvement [53]. We downgraded the certainty of evidence where concerns were identified. We applied the methods and recommendations outlined in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [53], and we used the GRADEpro GDT software to guide our assessments.

Consumer involvement

Consumers assessed the final draft of the review and provided feedback on the plain language summary. We also ranked and used items from a core outcome set for the review's outcomes, which were developed with consumer involvement [36].

RESULTS

Description of studies

Table 1 is a summary of included studies and outcomes. For further information on included studies, excluded studies, studies awaiting classification and ongoing studies, refer to [Supplementary material 2](#), [Supplementary material 3](#), [Supplementary material 4](#) and [Supplementary material 5](#).

Results of the search

For this update, we identified 34,520 unique records through our electronic search. In assessing the studies, we used Cochrane's Screen4Me workflow to help identify potential reports of randomised trials in searches conducted in 2021 and 2022. The results of the Screen4Me assessment process can be seen in [Figure 1](#). We assessed the remaining records left after the Screen4Me process for 2021 ([Figure 2](#)) and 2022 ([Figure 3](#)). The search results from 2023 and 2025 (Screen4Me was not used for these searches) were imported into Covidence for screening. After the removal of duplicates, we screened 20,425 titles and abstracts for eligibility. We identified 11 studies for full-text review. We considered six citations to be relevant after reviewing the full text (ChiCTR1900025141; Chin 2018; Hosseinzadeh 2014 [54]; Li 2021; Lin 2021 [55]; Lu 2021 [56]). We retrieved the full text of five studies included in this update (Chin 2018; Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021). We excluded three full-text studies in this update (Johann 2016 [57]; Kang 2021 [58]; Wang 2014 [59]), for a total of 12 studies (Arnold 1977 [60]; Cobb 1992 [61]; Eyer 1990 [62]; Haddad 2006 [63]; Kerin 1991 [64]; May 1996 [65]; Nakae 2010 [66]; Panadero 2002 [67]; Rijnders 2004 [68]). The remaining study was identified by searching clinical trial registries (ChiCTR1900025141). We contacted the study author and were unable to obtain outcome data. However, we were advised that this has been submitted for publication; therefore, this study is ongoing ([Supplementary material 5](#)). One remaining study is awaiting classification from our prior update (ChiCTR-IOR-17010879). We have been unable to reach the study authors.

Figure 2. Screen4Me flow diagram - 2021

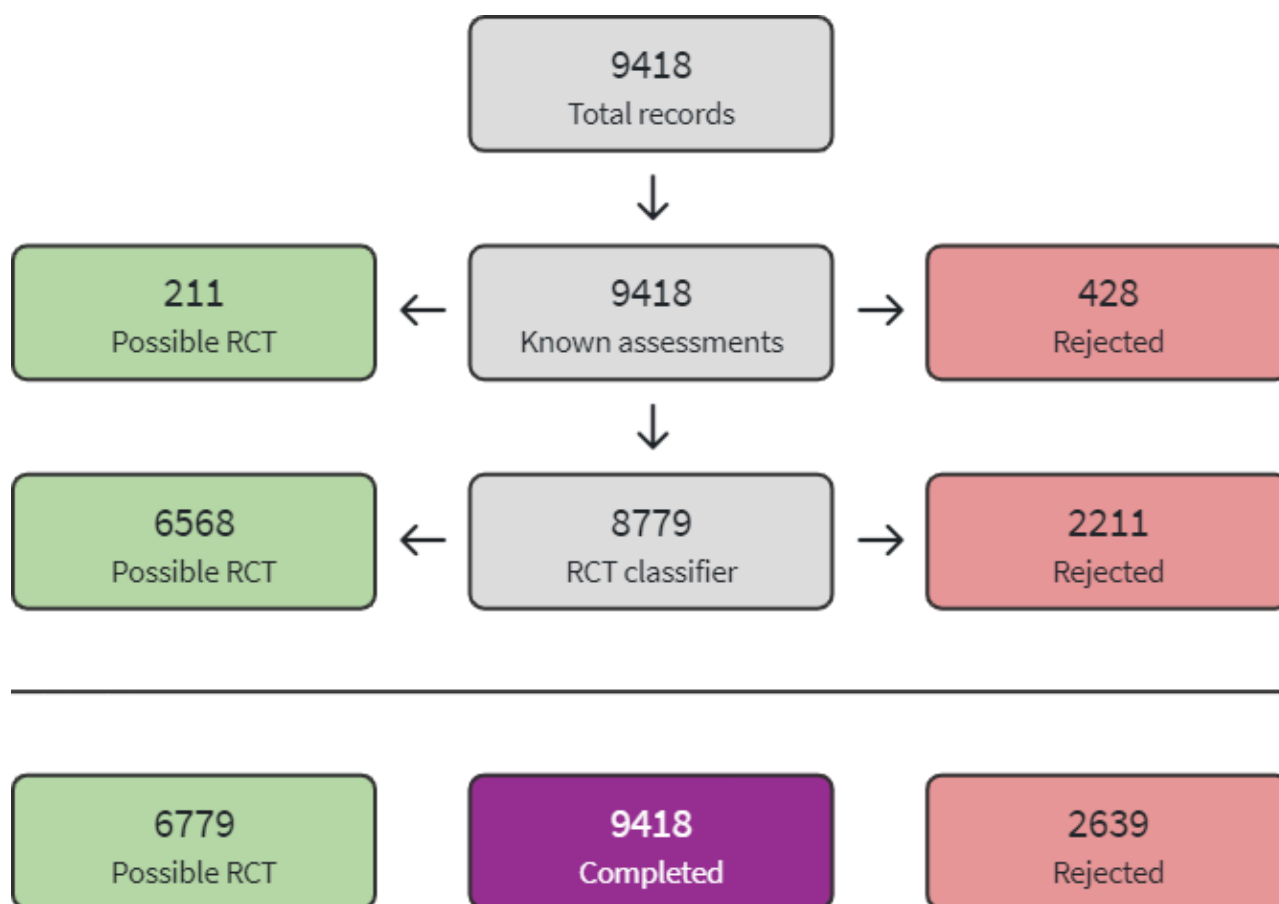
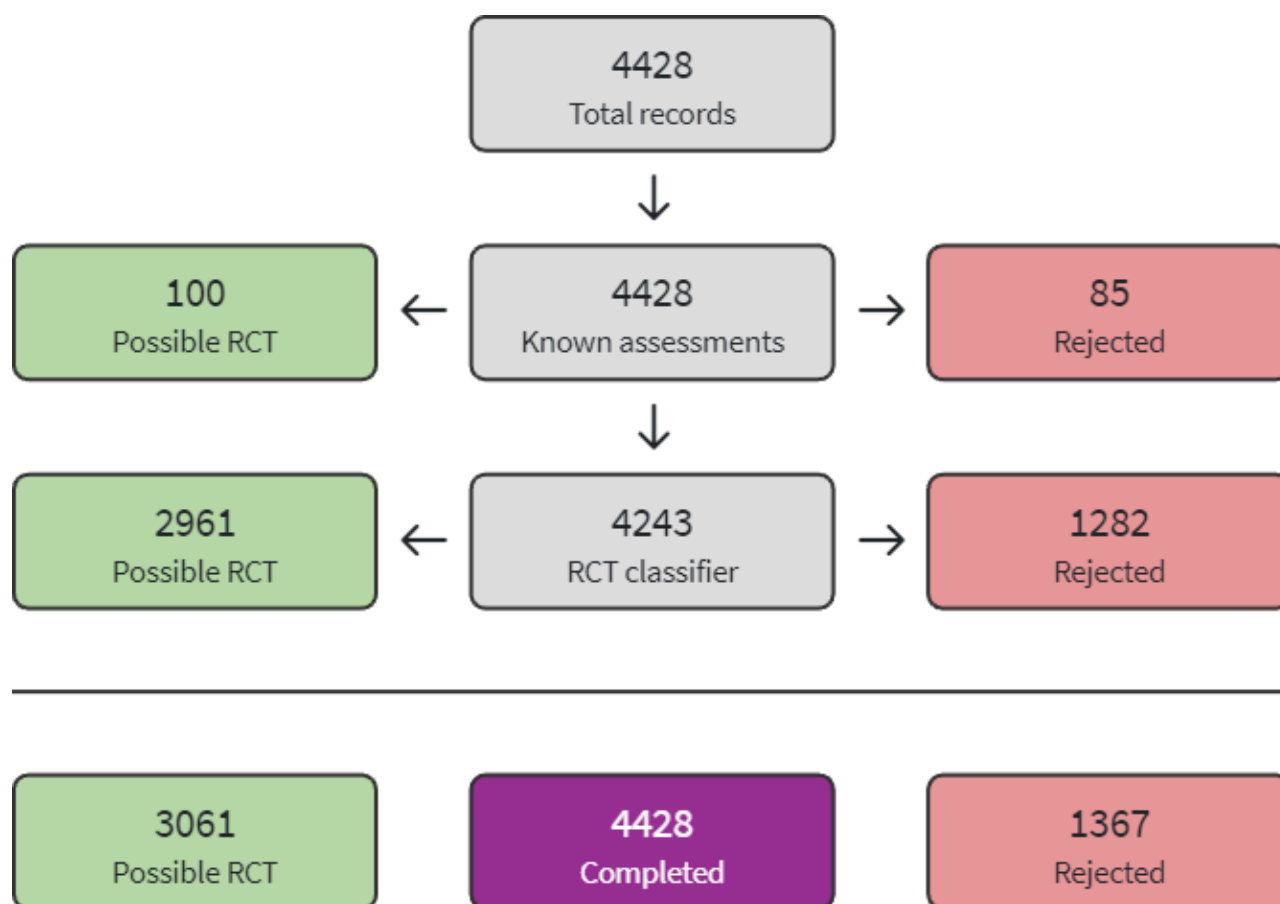


Figure 3. Screen4Me flow diagram - 2022



Our screening process is shown in [Figure 1](#).

No post-publication amendments were identified for any of the included or eligible studies.

Included studies

Characteristics of studies

All 14 included studies were RCTs, involving a total of 11,428 participants (Barker 2004 [69]; Chin 2018; Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021; Nishanth 2009 [70]; Rickard 2010 [71]; Rickard 2012 [72, 73, 74]; Van Donk 2009; Vendramim 2020 [75, 76]; Webster 2007 [77]; Webster 2008 [78]; Xu 2017). One study was a cluster trial, which randomised 20 hospital wards (Xu 2017). Sample sizes ranged from 42 (Nishanth 2009) to 3283 (Rickard 2012) participants.

Six studies were conducted in Australia (Chin 2018; Rickard 2010; Rickard 2012; Van Donk 2009; Webster 2007; Webster 2008), three in China (Li 2021; Lu 2021; Xu 2017) and one each in Brazil (Vendramim 2020), the United Kingdom (Barker 2004), India (Nishanth 2009), Iran (Hosseinzadeh 2014) and Taiwan (Lin 2021). There were equal proportions of LMICs (Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021; Nishanth 2009; Vendramim 2020; Xu 2017) and HICs (Barker 2004; Chin 2018; Rickard 2010; Rickard 2012; Van Donk 2009; Webster 2007; Webster 2008) represented in the studies.

All studies had institutional ethical approval.

Characteristics of participants

Eleven studies recruited adult participants (Barker 2004; Li 2021; Lu 2021; Nishanth 2009; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017); the remaining studies were conducted in neonatal or paediatric populations (Chin 2018; Hosseinzadeh 2014; Lin 2021).

Most studies reported participants receiving a combination of continuous or intermittent infusions for intravenous medication therapy (Barker 2004; Chin 2018; Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021; Nishanth 2009; Rickard 2010; Rickard 2012; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). Only Van Donk 2009 reported catheters that were used for intermittent intravenous therapy only.

In some studies, it was clear that outcomes for all PVCs per participant were included, but in several studies, it was unclear whether only the first PVC was reported (e.g. Xu 2017 had much longer follow-up for clinically indicated participants, suggesting not all PVCs used per participant in the routine removal group were reported).

Characteristics of interventions

In 12 studies, the routine removal interval was 72 to 96 hours (Chin 2018; Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). In most studies, reasons for clinically indicated removal were not clearly defined. Barker 2004 and Nishanth 2009 involved 48-hour routine removal and described clinically indicated removal reasons as including pain, catheter dislodgement or thrombophlebitis. Bedside clinicians primarily delivered interventions, not research or IV teams.

Ongoing studies

One study was identified through searches of clinical trial registries (ChiCTR1900025141). We contacted the study author for further information but were unable to obtain outcome data at the time of publication. This study remains ongoing ([Supplementary material 5](#)).

Studies awaiting classification

One study identified in the previous update remains awaiting classification (ChiCTR-IOR-17010879). We have been unable to contact the study authors to obtain additional information required for inclusion ([Supplementary material 4](#)).

Excluded studies

We excluded three new studies in this update (Johann 2016; Kang 2021; Wang 2014). [Supplementary material 3](#) contains a summary of the exclusion rationale. In summary, Johann 2016 and Wang 2014 did not compare routine removal with clinically indicated removal and Kang 2021 conducted a cross-over, cluster-RCT. [Supplementary material 3](#) also provides the rationale for excluding nine studies from the previous update (Arnold 1977; Cobb 1992; Eyer 1990; Haddad 2006; Kerin 1991; May 1996; Nakae 2010; Panadero 2002; Rijnders 2004). Two of these were small studies involving the administration of peripheral parenteral nutrition, and they did not compare straightforward routine replacement with clinically indicated removal (Kerin 1991; May 1996). Panadero 2002 compared one group that used the same catheter intraoperatively and postoperatively with a group using two catheters. Haddad 2006 compared 72-hour changes with 96-hour changes, and Cobb 1992, Eyer 1990, Nakae 2010 and Rijnders 2004 involved central venous catheters. The remaining study excluded was not an RCT (Arnold 1977).

Risk of bias in included studies

See the individual risk of bias judgements within the 'Characteristics of included studies' table ([Supplementary material 2](#)) and the summary figures ([Figure 4](#); [Figure 5](#)).

Figure 4. 'Risk of bias' graph: review authors' judgements about each risk of bias item presented as percentages across all included studies

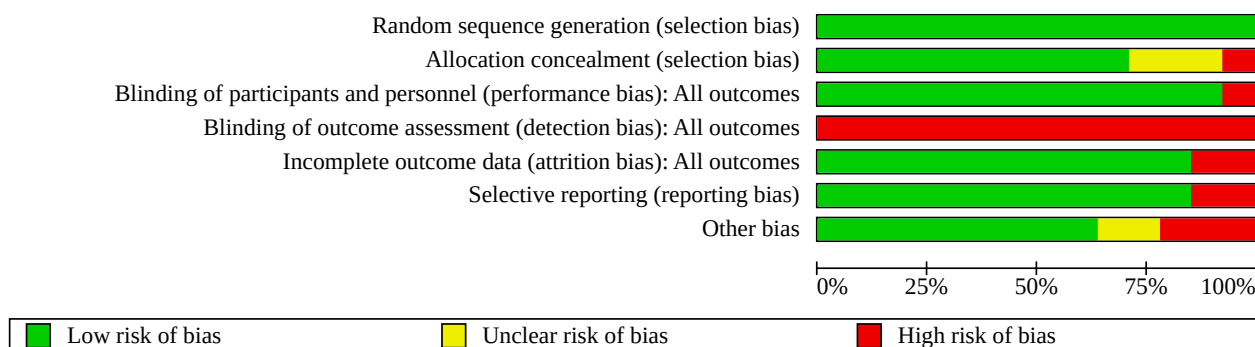


Figure 5. 'Risk of bias' summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Barker 2004	+	+	-	-	-	+	-
Chin 2018	+	+	+	-	+	-	?
Hosseinzadeh 2014	+	?	+	-	+	+	+
Li 2021	+	+	+	-	+	+	+
Lin 2021	+	?	+	-	+	+	+
Lu 2021	+	?	+	-	-	-	+
Nishanth 2009	+	-	+	-	+	+	-
Rickard 2010	+	+	+	-	+	+	?
Rickard 2012	+	+	+	-	+	+	+
Van Donk 2009	+	+	+	-	+	+	+
Vendramim 2020	+	+	+	-	+	+	+
Webster 2007	+	+	+	-	+	+	+
Webster 2008	+	+	+	-	+	+	+
Xu 2017	+	+	+	-	+	+	-

Selection bias

Generation of random allocation sequence

Most investigators (11 out of 14) used a computer-based sequence generator for random allocation, with one study employing a 'coin toss' and two using a random number table. All were judged to be at low risk of bias.

Allocation concealment

We deemed the majority of studies (10 out of 14) at low risk of bias for appropriate allocation concealment methods, while one study was at high risk due to investigator involvement (Nishanth 2009). We labelled three studies as at unclear risk of bias due to insufficient reporting (Hosseinazadeh 2014; Lin 2021; Lu 2021).

Performance bias: blinding of participants and personnel

Blinding the removal policy to participants or hospital staff was not feasible; however, we considered that the awareness of participants and personnel of the allocation would not likely have influenced the outcome. Therefore, we considered studies at low risk of performance bias. We deemed one study at high risk due to the investigator's involvement in all study operationalisation (Barker 2004).

Detection bias: blinding of outcome assessment

Overall, we considered all studies at high risk of detection bias in this domain, as all included unblinded clinical staff and participants. However, in some studies, CRBSI and primary BSI determination were assessed by laboratory staff who were blinded to group allocation (Li 2021; Rickard 2010; Rickard 2012; Webster 2008; Xu 2017). In two studies it was unclear whether laboratory staff were unblinded (Hosseinazadeh 2014; Lu 2021), while three studies were conducted without any blinding for CRBSI and primary BSI outcomes (Van Donk 2009; Vendramim 2020; Webster 2007). Blinding of outcome assessment was not feasible for subjective outcomes (e.g. patient-reported pain, phlebitis and infiltration).

Attrition bias: incomplete outcome data

We assigned high-risk judgements to studies with incomplete reporting and/or unbalanced participant numbers between groups. We considered most studies at low risk, providing complete outcome data and participant flow throughout the study. We judged two studies at high risk due to a lack of CONSORT flow charts and unbalanced participant numbers (Barker 2004; Lu 2021).

Reporting bias: selective outcome reporting

We assessed 12 studies as at low risk of reporting bias, with nine studies registered prospectively or retrospectively on clinical trial

registries (Hosseinazadeh 2014; Li 2021; Rickard 2010; Rickard 2012; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017) and four unregistered but reporting all prespecified outcomes (Barker 2004; Lin 2021; Nishanth 2009; Van Donk 2009). We deemed two studies at high risk for selective reporting due to incomplete reporting, or discrepancies between registered and reported outcomes (Chin 2018; Lu 2021). Where discrepancies existed, we contacted trial authors for clarification.

Other bias

Eight studies were at low risk of other bias (Hosseinazadeh 2014; Li 2021; Lin 2021; Lu 2021; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2008; Xu 2017), two were unclear (Chin 2018; Rickard 2010) and three were considered high risk (Barker 2004; Nishanth 2009; Xu 2017). Other biases included extreme results (Nishanth 2009), baseline group differences in participants (Chin 2018; Rickard 2010), trial design (cluster-RCT) and significant protocol deviations (Xu 2017), and research investigator involvement in study operationalisation (Barker 2004).

Synthesis of results

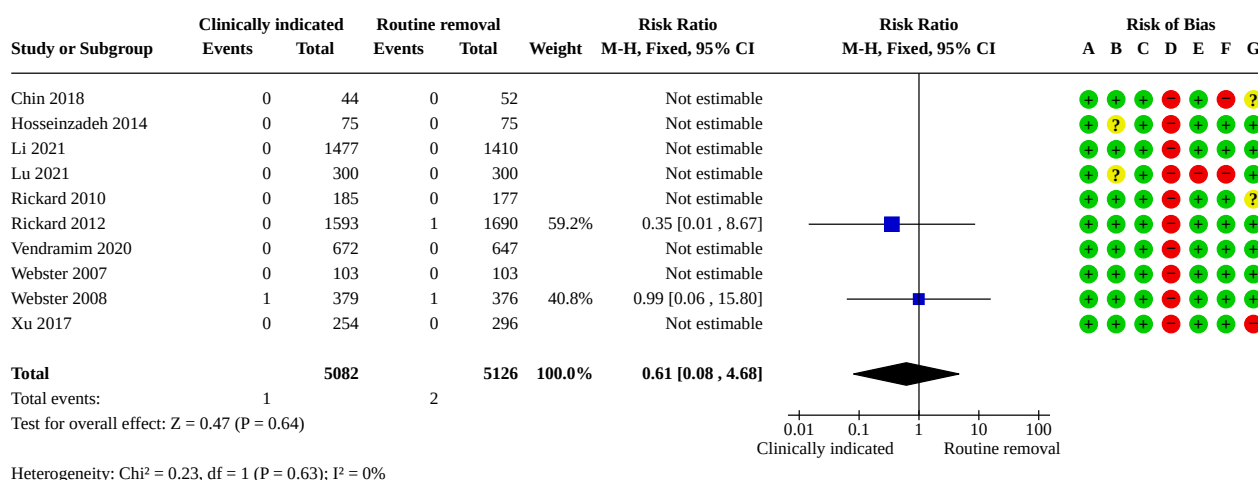
A complete record of all analyses conducted for this review and the comprehensive data package are provided in [Supplementary material 6](#), [Supplementary material 7](#) and [Supplementary material 9](#). [Summary of findings 1](#) summarises the absolute and relative effects for all critical and important outcomes, accompanied by their corresponding GRADE ratings indicating the level of confidence in the effect estimates.

Critical outcomes

Catheter-related bloodstream infection

Ten studies (10,208 participants) assessed catheter-related bloodstream infection (CRBSI) (Chin 2018; Hosseinazadeh 2014; Li 2021; Lu 2021; Rickard 2010; Rickard 2012; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017), but no CRBSI events were reported in eight studies (Chin 2018; Hosseinazadeh 2014; Li 2021; Lu 2021; Rickard 2010; Vendramim 2020; Webster 2007; Xu 2017). Clinically indicated removal compared with routine removal likely results in little to no difference in catheter-related bloodstream infection (CRBSI) (risk ratio (RR) 0.61, 95% confidence interval (CI) 0.08 to 4.68; fixed-effect; $I^2 = 0\%$; 10 studies, 10,208 participants; [Figure 6](#)). See Analysis 1.1 in [Supplementary material 6](#). We downgraded the certainty of the evidence by two levels to low, due to very serious imprecision (few events and wide confidence intervals), creating uncertainty around the estimate ([Summary of findings 1](#)). There were insufficient data to complete subgroup or sensitivity analyses.

Figure 6. This figure shows the pooled effect estimate for the risk of catheter-related bloodstream infection comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Risk of bias legend

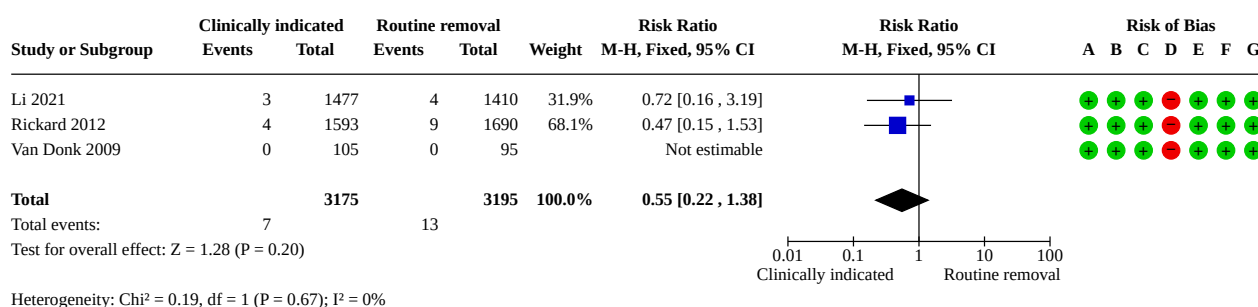
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Primary bloodstream infection

Three studies assessed primary bloodstream infection (Li 2021; Rickard 2012; Van Donk 2009). Clinically indicated removal compared with routine removal likely results in little to no difference in primary bloodstream infection (RR 0.55, 95% CI 0.22

to 1.38; fixed-effect; I² = 0%; 3 studies, 6370 participants; Figure 7). See Analysis 1.2 in [Supplementary material 6](#). We downgraded the evidence to low certainty, downgrading by two levels due to very serious imprecision (few events and wide confidence intervals). There were insufficient data to complete subgroup or sensitivity analyses.

Figure 7. This figure shows the pooled effect estimate for the risk of primary bloodstream infection comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Risk of bias legend

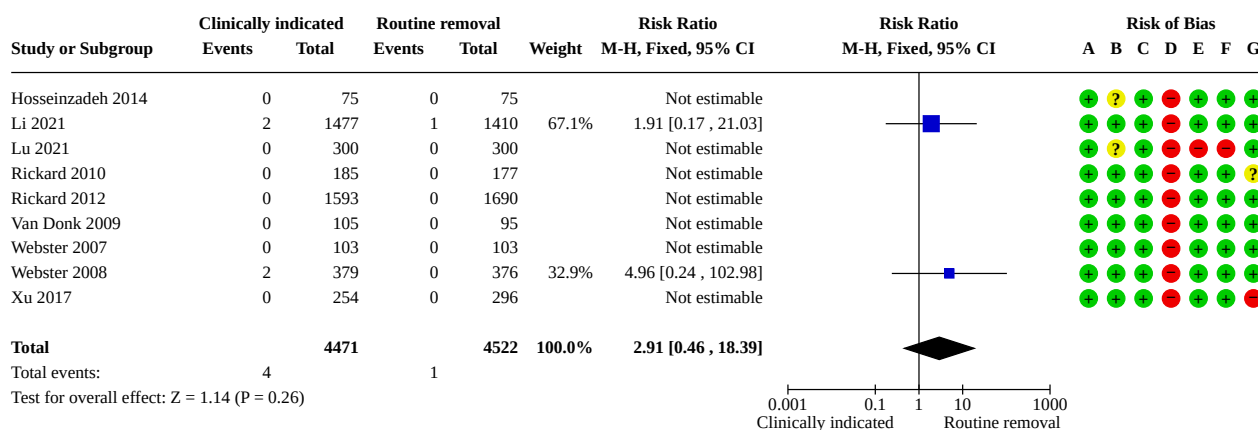
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Local infection

Nine studies measured local infection (Hosseinzadeh 2014; Li 2021; Lu 2021; Rickard 2010; Rickard 2012; Van Donk 2009; Webster 2007; Webster 2008; Xu 2017). Clinically indicated removal compared with routine removal may result in little to no difference in local infection

(RR 2.91, 95% CI 0.46 to 18.39; fixed-effect; $I^2 = 0\%$; 9 studies, 8993 participants; [Figure 8](#)). See Analysis 1.3 in [Supplementary material 6](#). We downgraded the evidence to low certainty due to very serious imprecision (few events and wide confidence intervals) ([Summary of findings 1](#)). There were insufficient data to complete any subgroup or sensitivity analyses.

Figure 8. This figure shows the pooled effect estimate for the risk of local infection comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Risk of bias legend

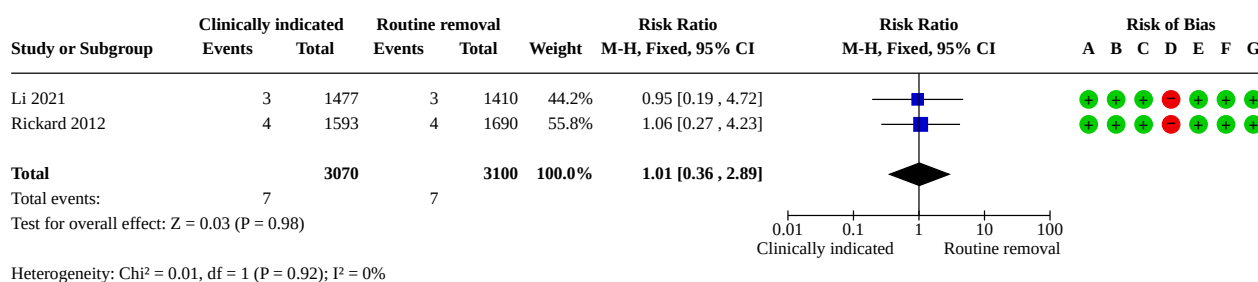
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Mortality

Two studies reported mortality (Li 2021; Rickard 2012). Clinically indicated removal compared with routine removal may result in little to no difference in mortality (RR 1.01, 95% CI 0.36 to 2.89; fixed-effect; $I^2 = 0\%$; 2 studies, 6170 participants; [Figure 9](#)). See

Analysis 1.4 in [Supplementary material 6](#). We downgraded the certainty of evidence by two levels to low certainty due to very serious imprecision (few events and wide confidence intervals) ([Summary of findings 1](#)). There were insufficient data to complete any subgroup or sensitivity analyses.

Figure 9. This figure shows the pooled effect estimate for the risk of mortality comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

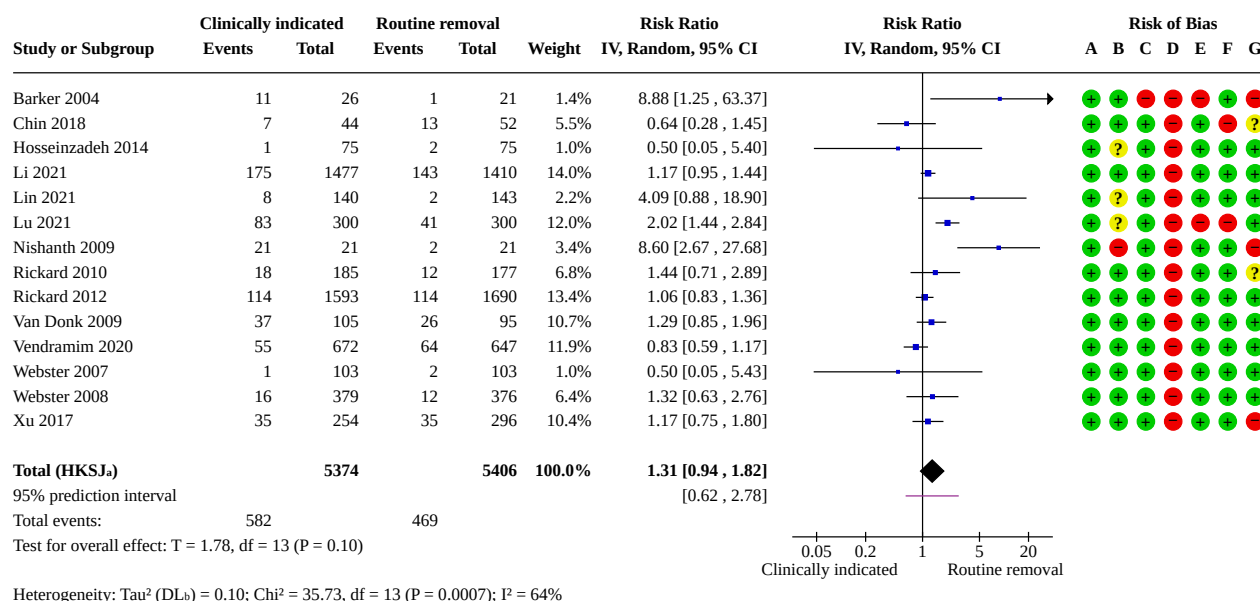
Important outcomes

Thrombophlebitis

All 14 studies reported thrombophlebitis (Barker 2004; Chin 2018; Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021; Nishanth 2009; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). There were 1051 cases of thrombophlebitis in 10,780 participants (9.7%). Clinically indicated removal of PVCs may increase the risk of thrombophlebitis, but the evidence is very uncertain (RR 1.31, 95% CI 0.94 to 1.82; random-

effects; $I^2 = 64\%$; 14 studies, 10,780 participants; Figure 10). See Analysis 1.5 in [Supplementary material 6](#). We downgraded the certainty of the evidence by three levels to very low certainty. We downgraded due to the overall risk of bias, as three of the 14 studies were assessed to be at high risk across three or more bias domains (Barker 2004; Lu 2021; Nishanth 2009), inconsistency (substantial clinical heterogeneity) and imprecision (95% CI crosses both the null effect (1.0) and multiple thresholds of clinical significance), which affected the certainty of the evidence.

Figure 10. This figure shows the pooled effect estimate for the risk of thrombophlebitis comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Footnotes

^aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

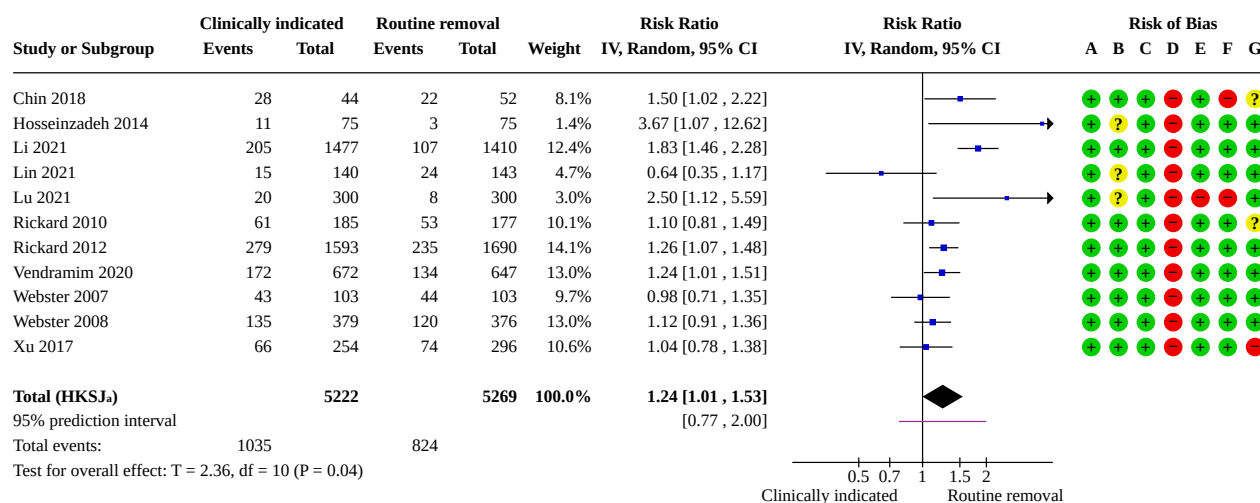
The effect showed no evidence of an outcome difference after applying a priori sensitivity analyses. The difference between interventions was less pronounced in post hoc sensitivity analysis based on the exclusion of studies reporting outcomes for single catheters (RR 1.09, 95% CI 0.88 to 1.37; 9 RCTs, 9155 participants). There was no statistical or visible difference in subgroup analysis for age differences, HIC versus LMIC, or for continuous or intermittent infusions. We performed a subgroup analysis to test whether routine removal intervals modify the effect on thrombophlebitis. There was a statistically significant difference in effect estimates in the subgroups for routine removal intervals ($P < 0.00001$), with the clinically indicated removal showing a higher risk of thrombophlebitis compared with the 48-hour routine removal subgroup (RR 8.67, 95% CI 7.23 to 10.40; random-effects). However, there were a small number of studies and participants in the 48-hour routine removal group (2 studies, 89 participants) and 72- to 96-hour routine removal group (2 studies, 296 participants) with moderate unexplained heterogeneity between the studies within the 96-hour removal subgroups ($I^2 = 85\%$), meaning the analysis

is unlikely to produce useful findings (Supplementary material 6; Supplementary material 10).

Thrombophlebitis per 1000 catheter-days

Nine studies provided thrombophlebitis rates per catheter-day (Chin 2018; Li 2021; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). Clinically indicated removal compared with routine removal of PVCs likely results in little to no difference in the rate of thrombophlebitis per 1000 catheter-days (rate ratio (RaR) 0.98, 95% CI 0.78 to 1.22; random-effects; $I^2 = 36\%$; 9 studies, 9658 participants; Figure 11). See Analysis 1.6 in Supplementary material 6. We downgraded by two levels, once for serious inconsistency (moderate heterogeneity $I^2 = 36\%$) and once due to serious risk of bias (no blind outcome assessment in any of the studies). The results did not suggest a visible or statistical difference between the investigated subgroups (Supplementary material 10). There were insufficient data to conduct a sensitivity analysis for this outcome.

Figure 12. This figure shows the pooled effect estimate for the risk of infiltration comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Footnotes

^aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

There was no clinically significant difference in the subgroup analysis for age differences, or HIC versus LMIC. We performed a subgroup analysis to test whether routine removal intervals modify the effect of infiltration (analysis not presented). There was a statistically significant difference in effect estimates in the subgroups for routine removal intervals ($P = 0.08$), with the 72- to 96-hour subgroup showing a higher risk of infiltration with clinically indicated removal compared with routine removal (RR 1.50, 95% CI 1.02 to 2.22; random-effects). However, there were a small number of studies and participants in the 72- to 96-hour routine removal group (1 study, 96 participants) compared to the 72-hour (7 studies, 5589 participants) and 96-hour routine removal subgroups (3 studies, 4806 participants). There was also moderate unexplained heterogeneity between the studies within the 96-hour removal subgroups ($I^2 = 76\%$), indicating that the analysis is unlikely to produce useful findings. No significant changes were observed in the pooled effect estimates or the direction of effect across both a priori and post hoc sensitivity analyses (Supplementary material 6; Supplementary material 11).

Infiltration per 1000 catheter-days

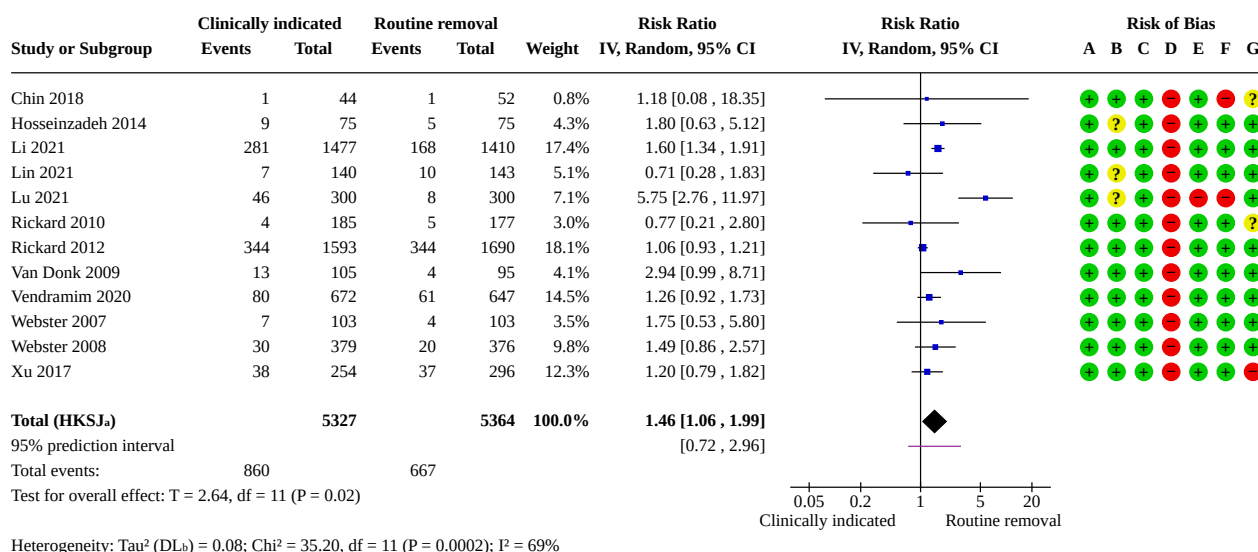
We compared available infiltration rates per 1000 catheter-days in eight studies (Chin 2018; Li 2021; Rickard 2010; Rickard 2012; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). Clinically indicated removal compared with routine removal likely results

in little to no difference in infiltration rate per 1000 catheter-days (RaR 1.11, 95% CI 0.79 to 1.57; random-effects; $I^2 = 85\%$; 8 studies, 9458 participants). See Analysis 1.8 in Supplementary material 6. We downgraded by two levels to low certainty, once for serious inconsistency (considerable heterogeneity $I^2 = 85\%$) and once for serious risk of bias (no blinded outcome assessment in any of the studies). There were no clinically significant differences in the subgroup analysis for age groups, HIC versus LMIC, or routine removal frequency (Supplementary material 11). There were insufficient data to conduct sensitivity analyses for this outcome.

Catheter occlusion

Twelve studies assessed catheter occlusion (Chin 2018; Hosseinzadeh 2014; Li 2021; Lin 2021; Lu 2021; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). Clinically indicated removal compared with routine removal of PVCs results in an increase in catheter occlusion (RR 1.46, 95% CI 1.06 to 1.99; random-effects; $I^2 = 69\%$; 12 studies, 10,691 participants; Figure 13). See Analysis 1.9 in Supplementary material 6. We downgraded the certainty of the evidence by two levels, once for serious risk of bias (no blinded outcome assessors) and once for serious inconsistency (substantial heterogeneity $I^2 = 69\%$) (Summary of findings 1).

Figure 13. This figure shows the pooled effect estimate for the risk of occlusion comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Risk ratios (RR) and 95% confidence intervals (CIs) are presented.



Footnotes

^aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

No significant changes were observed in the pooled effect estimates or the direction of effect across a priori sensitivity analyses. The difference between interventions was less pronounced in post hoc sensitivity analysis based on exclusion of studies reporting outcomes for single catheters (RR 1.18, 95% CI 0.76 to 1.82; 7 RCTs, 7709 participants). There was no clinically significant difference in subgroup analysis for age differences or HIC versus LMIC. We performed a subgroup analysis to test whether routine removal intervals modify the effect of catheter occlusion. There was a statistically significant difference in effect estimates in the subgroups for routine removal intervals ($P = 0.01$), with the 72-hour subgroup showing a higher risk of catheter occlusion with clinically indicated removal compared with routine removal (RR 1.09, 95% CI 0.96 to 1.24; random-effects). However, there were a small number of studies and participants in the 72- to 96-hour routine removal group (1 study, 296 participants) compared to the 72-hour (7 studies, 5589 participants) and 96-hour routine removal subgroups (3 studies, 4806 participants). There was also moderate unexplained heterogeneity between the studies within the 96-hour removal subgroups ($I^2 = 86\%$), indicating that the analysis is unlikely to produce useful findings (Supplementary material 6; Supplementary material 12).

Catheter occlusion per 1000 catheter-days

We compared available occlusion rates per 1000 catheter-days in nine studies (Chin 2018; Li 2021; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2007; Webster 2008; Xu 2017). Clinically indicated removal compared with routine removal likely results in little to no difference in the rate of catheter occlusion per 1000 catheter-days (RaR 1.18, 95% CI 0.85 to 1.63; random-effects; $I^2 = 75\%$; 9 studies, 9658 participants). See Analysis 1.10 in Supplementary material 6. We downgraded by two levels to low certainty, once for serious inconsistency (substantial inconsistency $I^2 = 75\%$) and once for serious risk of bias (no blinded outcome assessment in any of the studies).

There was no clinically significant difference in subgroup analysis for age differences, continuous versus intermittent infusion, or HIC versus LMIC. We performed a subgroup analysis to test whether routine removal intervals modify the effect of catheter occlusion per 1000 catheter-days. The statistically significant difference in effect estimates between subgroups ($P = 0.001$) appears to be driven by variation in effect direction, with the 72-hour subgroup favouring clinically indicated replacement (RaR 0.93, 95% CI 0.61 to 1.42; random-effects) and the 72- to 96-hour and 96-hour subgroups favouring routine removal (RaR 2.23, 95% CI 0.19 to 25.59; random-effects; RaR 1.55, 95% CI 0.17 to 14.08; random-effects, respectively). However, the wide confidence intervals and

small number of studies within each subgroup limit the reliability of this finding ([Supplementary material 6](#); [Supplementary material 12](#)). There were insufficient data to conduct a sensitivity analysis for this outcome.

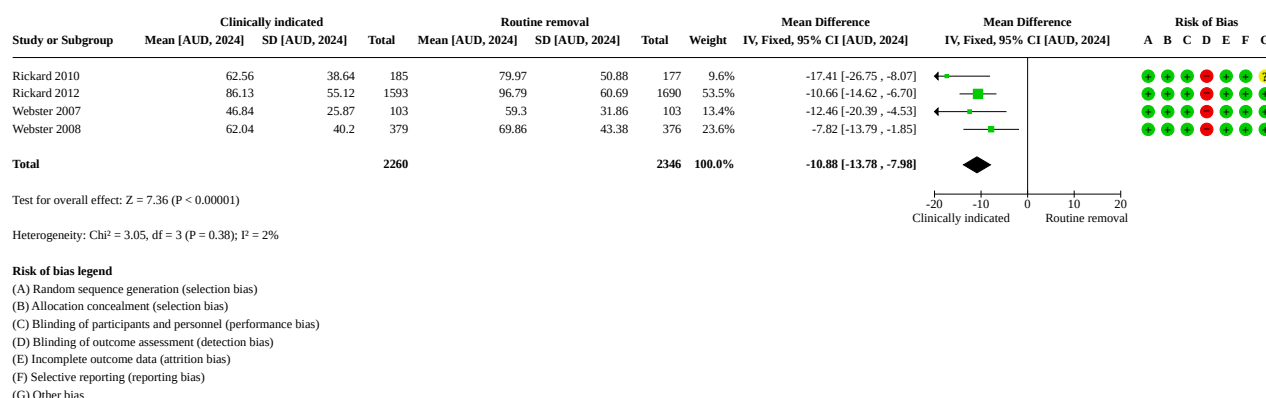
Limited importance outcomes

Cost

Five studies measured cost (Li 2021; Rickard 2010; Rickard 2012; Webster 2007; Webster 2008). Li 2021 was unable to provide cost

as mean and standard deviation (SD) but provided only median and interquartile range. Additionally, their cost calculation method differed, as it excluded staff time. Due to these differences, this study was not included in the meta-analysis. Device-related costs were measured in Australian dollars (AUD; 2024). Clinically indicated removal of PVCs was associated with an average cost reduction of approximately AUD 11.00 per participant compared with routine removal (mean difference (MD) -10.88, 95% CI -13.78 to -7.98; fixed-effect; $I^2 = 2\%$; 4 studies, 4606 participants; [Figure 14](#)). See Analysis 1.11 in [Supplementary material 6](#).

Figure 14. This figure shows the pooled effect estimate for the mean difference in costs comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Mean difference (MD) and 95% confidence intervals (CIs) are presented. [AUD: Australian dollars]



Pain during infusion

Only Vendramim 2020 reported pain using a validated pain assessment scale (range 1 to 10). There was no clear difference in device-related pain between the clinically indicated and routine removal group. The mean pain score in the clinically indicated group was 4.6 (SD 3.0) compared with 5.2 (SD 3.0) in the routine removal group (MD -0.60, 95% CI -1.44 to 0.24). See Analysis 1.12 in [Supplementary material 6](#).

Satisfaction

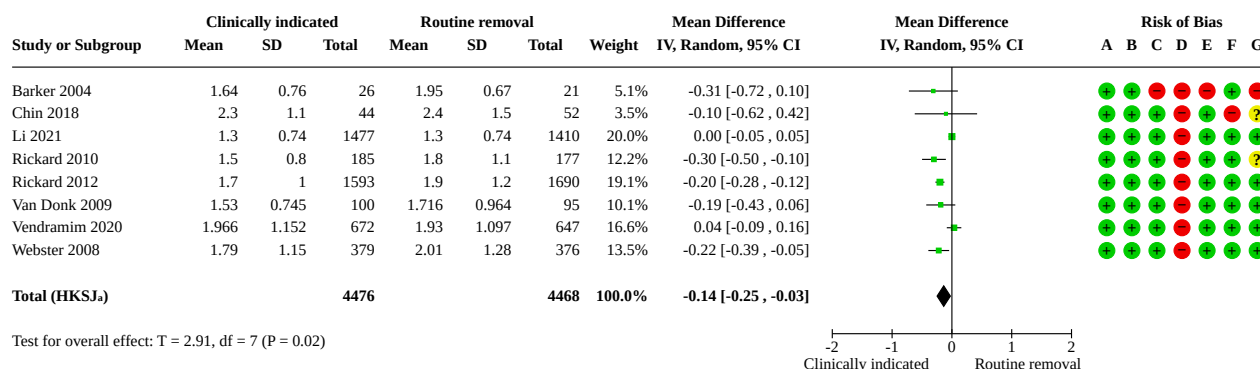
Vendramim 2020 developed and validated a patient satisfaction tool to assess the experience and satisfaction of patients with PVCs (range 0 to 75). The authors reported satisfaction results only for a subset ($n = 189$) of the overall cohort. There was no clear difference in satisfaction between clinically indicated removal and routine removal. The mean satisfaction score in the clinically indicated group was 56.6 (SD 11.6) compared with 55.3 (SD 11.5) in the routine

removal group (MD 1.3, 95% CI -2.00 to 4.60). See Analysis 1.13 in [Supplementary material 6](#).

Number of catheters per participant

Eight studies measured the number of catheters per participant for the total course of therapy in both study arms (Barker 2004; Chin 2018; Li 2021; Rickard 2010; Rickard 2012; Van Donk 2009; Vendramim 2020; Webster 2008). Nishanth 2009 did not follow participants for the complete duration of the therapy requirement in the routine removal arm therefore they were not included. Li 2021 did not provide catheter totals as mean and SD. We converted the reported median and interquartile range to mean and SD. Clinically indicated removal of PVCs was associated with a lower number of catheters used per participant compared with routine removal (MD -0.14, 95% CI -0.25 to -0.03; random-effects; $I^2 = 77\%$; 8 studies, 8944 participants; [Figure 15](#)). See Analysis 1.14 in [Supplementary material 6](#).

Figure 15. This figure shows the pooled effect estimate for the mean difference in the number of catheters per participant comparing clinically indicated replacement with routine replacement of peripheral venous catheters across included studies. Mean difference (MD) and 95% confidence intervals (CIs) are presented.



Footnotes

^aCI calculated by Hartung-Knapp-Sidik-Jonkman (HKSJ) method.

^b τ^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Equity assessment

There was no clinically significant difference in the subgroup analyses for HIC versus LMIC for the thrombophlebitis, infiltration or catheter occlusion outcomes.

Reporting biases

Funnel plots did not indicate a risk of publication bias for thrombophlebitis, infiltration or catheter occlusion (Figure 16; Figure 17; Figure 18).

Figure 16. This funnel plot assesses potential publication bias for studies comparing clinically indicated and routine replacement of peripheral venous catheters for thrombophlebitis. The vertical line represents the pooled effect estimate, and each study is plotted according to its effect size and standard error.

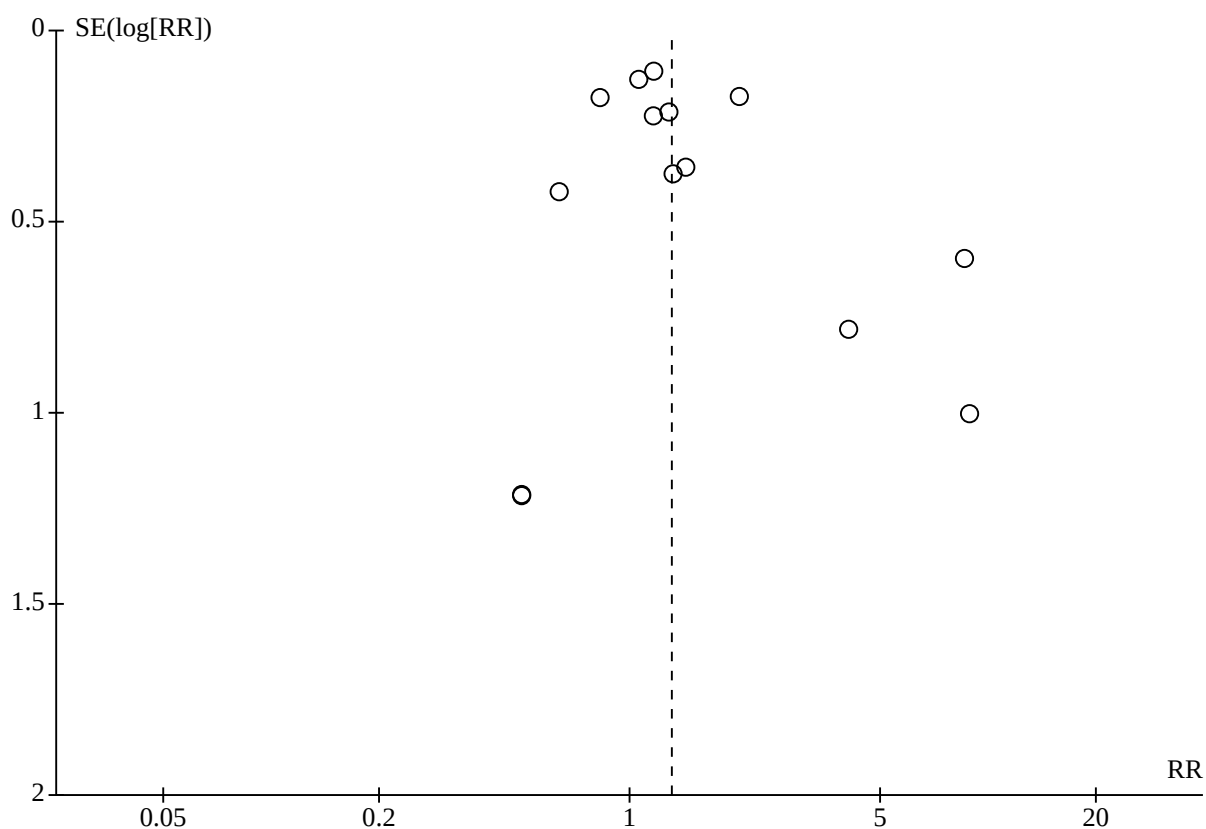


Figure 17. This funnel plot assesses potential publication bias for studies comparing clinically indicated and routine replacement of peripheral venous catheters for infiltration. The vertical line represents the pooled effect estimate, and each study is plotted according to its effect size and standard error.

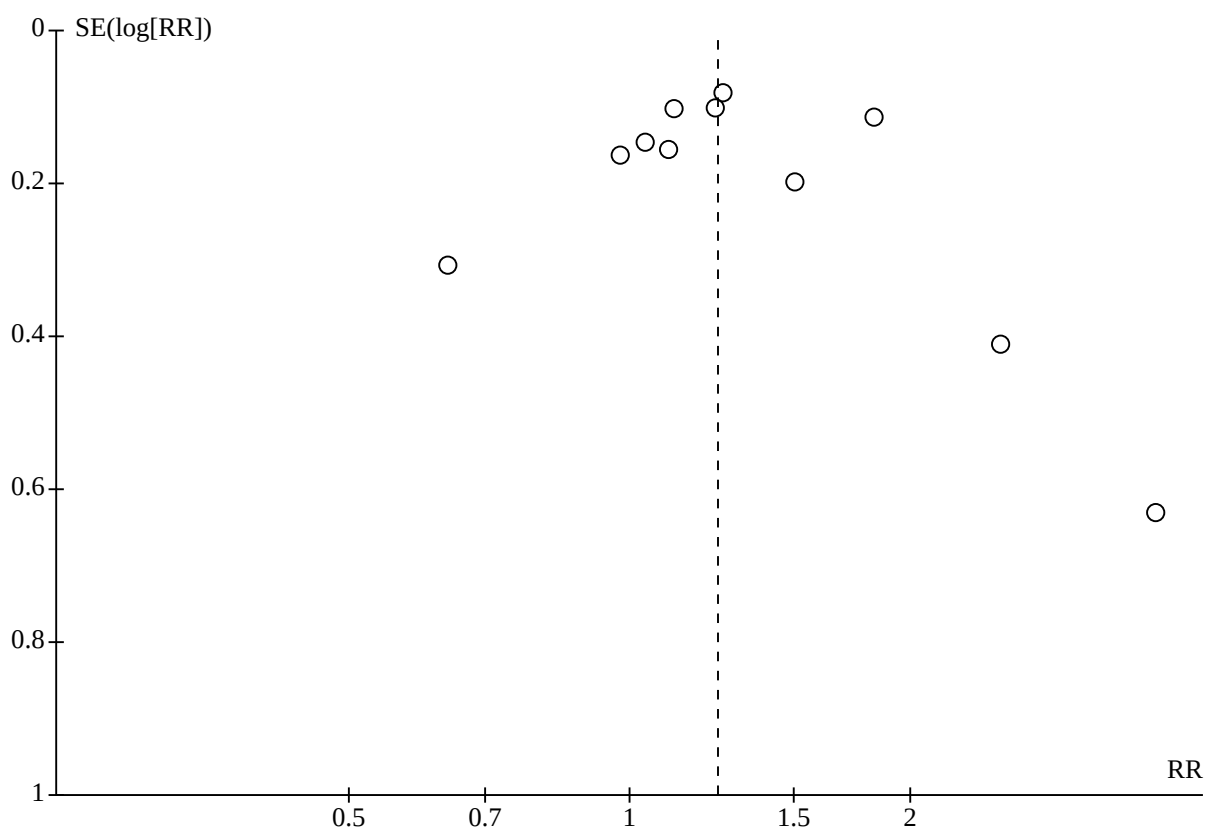
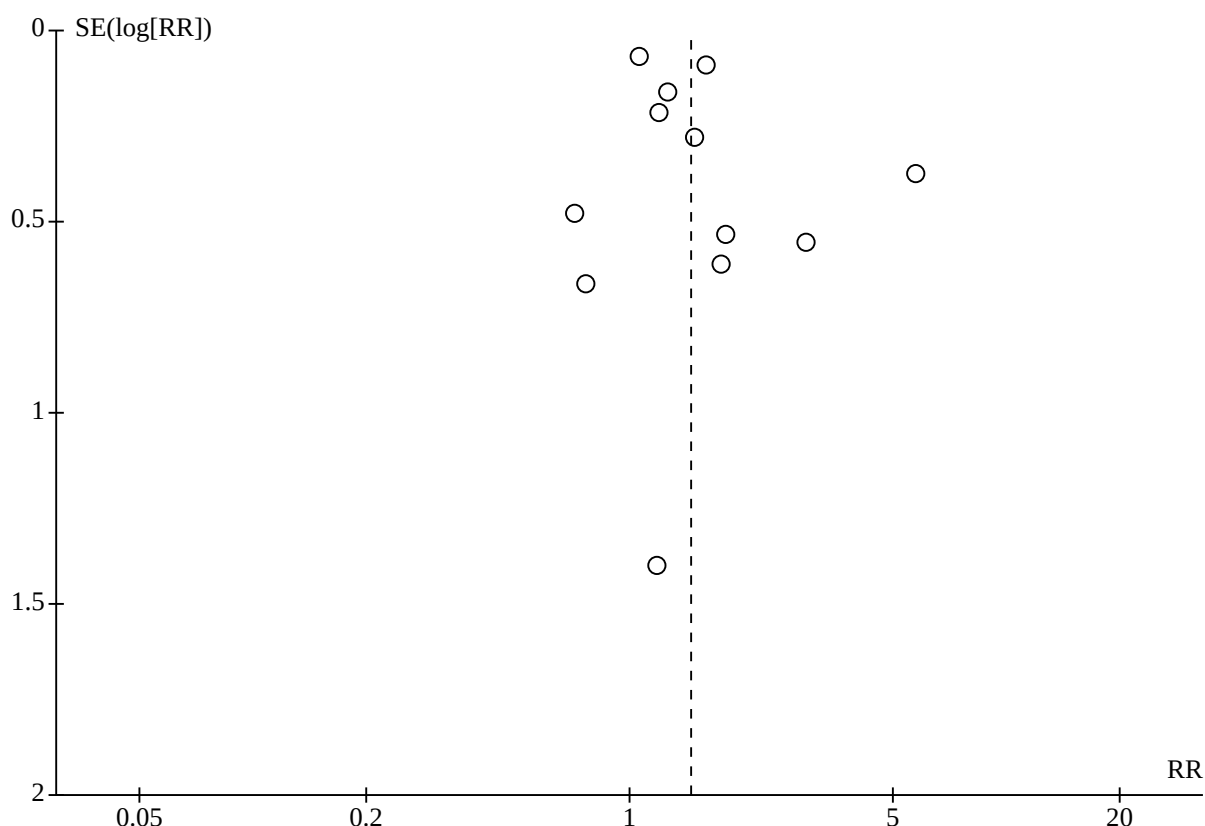


Figure 18. This funnel plot assesses potential publication bias for studies comparing clinically indicated and routine replacement of peripheral venous catheters for occlusion. The vertical line represents the pooled effect estimate, and each study is plotted according to its effect size and standard error.



DISCUSSION

Summary of main results

This review aimed to compare the effects of PVC removal timing (routine versus clinically indicated) on the incidence of catheter complications. The update included five new studies and an additional 4036 participants, bringing the total number of studies for inclusion to 14 with 11,428 participants. Outcomes were categorised as critical outcomes, important outcomes and outcomes of limited importance. Most studies reported a combination of outcomes with only thrombophlebitis reported by all studies.

Critical outcomes

The evidence suggests that uncertainty remains regarding clinically indicated removal of PVCs for infection-related outcomes, including CRBSI, primary BSI and local infection. These outcomes were rare in PVCs regardless of whether they were removed as clinically indicated or routinely. Compared to routine removal, clinically indicated removal of PVCs may result in little to no difference in mortality. More extensive studies are required for optimal information size for all infectious endpoints. However, an RCT with sufficient power to statistically test for differences in infection between PVCs removed routinely or by clinical indication would need to be very large; for example, around 25,000 patients

would be needed to test for differences in CRBSI. This conclusion remains unchanged from previous versions of the review.

Important outcomes

The evidence is very uncertain about the effect of clinically indicated removal on the outcome of thrombophlebitis. Clinically indicated removal of PVCs likely results in little to no difference in the rate of thrombophlebitis per 1000 catheter-days compared to routine removal. Subgroup analyses by removal schedules showed differential interventional effects for clinically indicated versus a 48-hour routine removal schedule on the incidence of thrombophlebitis per participant analysis. Post hoc sensitivity analyses showed a reduced effect estimate and narrower confidence interval for thrombophlebitis, indicating variability due to study design when studies reported outcomes in one or all PVCs per patient.

Evidence suggests that clinically indicated removal of PVCs may increase infiltration per participant, but likely results in little to no difference per 1000 days, compared to routine removal. Subgroup analyses by removal schedules showed differential interventional effects for clinically indicated versus a 96-hour routine removal schedule on the incidence of infiltration per participant. Sensitivity analyses (RCT quality, scientific integrity) supported the findings observed in the primary models.

Evidence suggests that clinically indicated removal of PVCs may increase catheter occlusion per participant, but likely results in little to no difference in the catheter occlusion rate per 1000 catheter-days compared to routine removal. Subgroup analysis for infiltration observed differential interventional effects for clinically indicated versus a 96-hour routine removal schedule on the incidence of catheter occlusion both per participant and per 1000 catheter-days. Post hoc sensitivity analyses showed a reduced effect estimate and wider confidence interval for catheter occlusion, indicating variability due to study design (one or multiple PVCs reported per patient).

Thrombophlebitis, infiltration and occlusion occurred frequently in PVCs whether removed as clinically indicated or routinely.

Limited importance

Clinically indicated removal likely reduces cost compared to routine removal. The cost was less, at around AUD 10.00 (2024) per participant, in the clinically indicated group. There was insufficient evidence to report pain and satisfaction; results were only reported in a single study. Evidence suggests that the clinically indicated removal of PVCs, compared to routine removal, likely results in fewer catheters required per participant for the duration of therapy. Because the overall rate of PVC complications was high in both groups, often necessitating additional PVC insertions, the resulting reduction in catheters per patient with clinically indicated removal was small.

Limitations of the evidence included in the review

Some results of our review should be interpreted with caution. [Summary of findings 1](#) outlines the factors that led to a downgrade in the certainty of the evidence using the GRADE approach. Overall, we found the certainty of the evidence for most outcomes to be of moderate to low certainty, primarily due to risk of bias (particularly non-blinding of outcome assessors) and imprecision. Imprecision was due in some cases, such as 'pain', to few participants, but in other cases, such as CRBSI, to few events. We assessed the risk of bias according to six components: sequence generation, allocation concealment, blinding, selective outcome reporting, incomplete follow-up and other potential biases. Four studies did not include information regarding allocation concealment to avoid selection bias (Hosseinzadeh 2014; Lin 2021; Lu 2021; Nishanth 2009).

We rated the overall methodological quality of Barker 2004, Lu 2021 and Nishanth 2009 as high risk of bias across multiple components, reducing confidence in their findings. The remaining studies were of moderate quality. Blinding was not possible for patients or clinical staff due to their direct involvement in catheter insertion and maintenance. This was considered unlikely to result in performance bias; however, we downgraded for detection bias, as the assessment of subjective outcomes, including phlebitis, infiltration and occlusion, was unblinded and therefore potentially susceptible to measurement bias. While its impact on outcomes is unclear, we downgraded by one level all outcomes (except for infectious outcomes) that could be assumed to be assessed by blinded microbiology staff. No studies used blinded assessors for thrombophlebitis, catheter occlusion or infiltration, likely as maintaining such oversight around the clock would be resource-intensive.

In all studies except for Barker 2004 and Nishanth 2009, control groups involved 72- to 96-hour PVC removal. Barker

2004 and Nishanth 2009 removed catheters every 48 hours. Only Rickard 2012 was powered to detect thrombophlebitis. The significantly lower thrombophlebitis rates in the routine removal groups of Barker 2004 and Nishanth 2009 (47 and 42 participants, respectively) contrast sharply with other studies, warranting cautious interpretation. Notably, Nishanth 2009 reported thrombophlebitis in all clinically indicated cases but none in the two-day change group – an extreme result unexplained by author correspondence. No other published studies reported a 100% thrombophlebitis rate.

The addition of five new studies since the last review update has introduced clinical heterogeneity in the population (age, country income group) and intervention (removal schedule, continuous versus intermittent) categories. Subgroup analyses showed no differences between age or infusion types. However, differences emerged in removal schedules and country income groups, though study variability and cross-pollination may have influenced these findings. Infiltration and catheter occlusion outcomes favoured routine removal in LMICs, potentially due to differences in resource availability, infection prevention strategies and training, which may contribute to higher PVC failure rates in some settings [9].

We noted design inconsistencies in whether complications for all PVCs per participant were reported. Lin 2021, Lu 2021 and Xu 2017 reported only one PVC per participant, which meant that fewer catheter days were studied to detect potential complications in the routine removal group than in the clinically indicated removal group. There was also inconsistency within studies, with Nishanth 2009 including only one PVC per participant in the clinically indicated removal group, but all PVCs per participant in the routine removal group. These design inconsistencies could be seen to influence variable estimates of effect seen between study outcomes per 100 participants and per 1000 PVC days, as well as in our post hoc sensitivity analysis.

Included studies reported intervention fidelity as an ongoing challenge. Studies reported large numbers of participants whose PVCs were not removed on time within the routine removal timeframe. For example, one-third of the routine removal participants in the study by Xu 2017 were removed on clinical indication due to poor veins or being forgotten. In addition, many studies saw the majority of clinically indicated removal participants having their PVCs replaced before the routine removal timeframe in the study, due to complications or treatment completion. For example, in Lu 2021, the average dwell was only 14 hours longer for clinically indicated removal PVCs (84 hours) than for those routinely removed (70 hours) and both were below the 96-hour routine removal time point. For these reasons, the reported average dwell time of PVCs in study groups was rarely different by more than a day, suggesting that routine removal may not be pragmatic and that other drivers of PVC complications are a higher priority for future research.

Limitations of the review processes

Although some review authors were investigators in one or more of the included studies, we followed clearly described procedures to prevent potential biases in the review process. We conducted a careful literature search, and our methods were transparent and reproducible. We feel confident that our comprehensive electronic searches, hand searching and searches of sizeable clinical trial databases identified all existing, published randomised controlled

trials addressing the review question. We completed funnel plot analysis per the protocol for thrombophlebitis, infiltration and catheter occlusion outcomes. We determined no publication bias at the time of the review publication.

Two potentially eligible trials are either awaiting classification or ongoing, due to limited information on trial governance and methodological compliance. We were advised that one has been submitted for publication. To minimise bias, two review authors assessed each trial's trustworthiness and contacted the authors for clarification in their native language. However, time constraints prevented the resolution of many concerns, leading to their exclusion from the current analyses. The omission of these potentially eligible studies may have influenced the review's findings.

Two review authors independently assessed RCTs for inclusion, trustworthiness and risk of bias, resolving disagreements through discussion. We evaluated the certainty of evidence using the GRADE approach. While these assessments involve some subjectivity and depend on trial reporting, we maintained consistency and transparency by documenting decisions and their rationale.

Some trial outcomes (cost and number of catheters per participant) were either not reported in publications or presented in a format unsuitable for analysis, and could not be incorporated into this review. The impact of the unavailable data may be substantial but is, at this stage, unknown.

Agreements and disagreements with other studies or reviews

Our review offers a current and thorough evaluation of the evidence. The results and conclusions of this systematic review corroborate and diverge from our previous findings [30] and those of other researchers [6, 19, 25, 79, 80].

A recent systematic review and meta-analysis by Chen et al. contains methodological inconsistencies that may have misrepresented the evidence, leading to recommendations favouring routine PVC removal [79]. The review lacked a transparent approach to handling missing or unclear data, did not indicate whether primary study authors were contacted and failed to explore heterogeneity or follow GRADE methodology. In updating our review, we sought clarification from several authors, resulting in differing incidence rates. We also included paediatric patients and studies assessing 48-hour routine removal. Logically, higher phlebitis, occlusion and infiltration rates per PVC are expected in the clinically indicated group if only one PVC is studied per participant due to more days at risk. Most PVCs will fail eventually, and complications are frequent even in the early days of dwell, before a routine removal policy would apply [9, 81]. Measuring days at risk, rather than incident rates alone, provides a more meaningful assessment for clinicians and policymakers.

One of our critical outcomes was CRBSI, assessed in 10 RCTs (10,906 participants). We found no evidence that dwell time influenced CRBSI; however, the results are imprecise due to the low overall incidence (clinically indicated: 0.02% (1/5007); routine removal: 0.03% (2/5051)). These rates align with a recent systematic review by Marsh et al. [6], which reported a pooled incidence of 0.028% across 38 studies. A systematic review, undertaken for the World Health Organization (WHO) to develop an infection prevention

guideline, meta-analysed six RCTs, finding that catheter-associated bloodstream infection (CABSI)/CRBSI associated with routine removal compared to clinically indicated removal led to a relative effect of RR 1.08 (95% credible interval 0.31 to 3.75) [82]. The WHO recommendation is consistent with our review, that "WHO suggests either the scheduled removal or clinically-indicated removal of PIVCs in adults, adolescents-children and neonates (Conditional recommendation; moderate certainty evidence" [82]). Adequately powered trials remain lacking, particularly for rare outcomes such as catheter-related bloodstream infection. Future research should consider large, cluster-randomised designs with CRBSI as the primary outcome, as recommended by the WHO guidelines [37]. We note that few trials provide clear definitions of what they considered clinically indicated removal criteria, and information was limited as to how staff were supported or audited as to whether decisions about removal were appropriate. Further research is needed in this area.

A recent secondary analysis of 12 pooled prospective studies by Marsh et al. (11,830 catheters) reported a 12% thrombophlebitis incidence, comparable to our findings in this review of 10% (clinically indicated: 10% (582/5374), routine removal: 9% (469/5406)) [24]. Previous studies reported that prolonged PVC dwell time did not increase thrombophlebitis risk, with most complications occurring in the first 96 hours [80, 83]. Observational studies (N = 38,161) [9] and (N = 469) [84] also found no increased risk with clinically indicated removal. Our meta-analysis of thrombophlebitis per 1000 catheter-days supports this finding. A systematic review and meta-analysis by the WHO included 10 RCTs reporting thrombophlebitis in adults and identified a relative risk for routine removal compared with clinically indicated removal of RR 0.74 (95% credible interval 0.49 to 1.01), with a high certainty of evidence that routine removal may result in a slight reduction in this outcome in adults, but very low certainty in children and neonates (for which only one RCT each was considered) [82]. In contrast, we are very uncertain about the impact of the removal approach for this outcome. This disagreement likely reflects our analysis of this outcome both per catheter and per 1000 catheter-days, whereas the WHO review considered only per PVC. In addition, the WHO did not consider the routine removal timeframes in included studies, whereas our sensitivity analyses identified that routine removal at 48-hour timeframes (two RCTs; Barker 2004; Nishanth 2009) had a significantly higher risk estimate for clinically indicated removal (RR 8.67, 95% CI 7.23 to 10.40). As the body takes approximately 48 hours to develop a detectable inflammatory response, this may explain why removing PVCs at 48 hours could reduce thrombophlebitis.

For the outcomes of infiltration and catheter occlusion, rates in routine and clinically indicated removal groups were higher than those reported in a 2020 systematic review [5]. Kaplan-Meier curves from a small study (N = 169) suggest that non-phlebitis catheter failures occur within 96 hours post-insertion [80], potentially limiting the benefits of routine removal. Any differential benefits of removal policies may not be realised if the time to event is similar for both clinically indicated and routine removal for catheter occlusion and infiltration. Clinical heterogeneity – due to differences in healthcare resources (HICs versus LMICs) and control group timing (48-, 72- or 96-hour routine removal) – introduces uncertainty around the estimate of effect, requiring cautious interpretation of results.

AUTHORS' CONCLUSIONS

Implications for practice

Evidence from randomised controlled trials assessing patients requiring peripheral venous catheters (PVCs) for medical treatments lasting over three days shows mixed results across multiple outcomes, with varying levels of certainty regarding optimal removal schedules. Healthcare organisations contemplating policy changes should carefully evaluate the risks and benefits specific to their settings. As the treatment required for thrombophlebitis, occlusion and infiltration is almost always limited to PVC removal and replacement, if therapy is ongoing, there is questionable benefit in routinely removing PVCs to avoid a potential increase in two to four cases of these complications per 100 PVCs, which is our current effect estimate. Since both the 'prevention' and 'treatment' interventions (PVC removal) are identical, it is hard to see a benefit in early removal. One practical application of these results is to inform patient education and involvement of informed patients in removal decisions, something that we do not believe happens often, but should. Transitioning towards clinically indicated removal offers potential advantages such as significant cost savings and improved patient comfort, reducing unnecessary re-sites when clinical indications are absent. Moreover, it could streamline clinical operations by reducing time spent on routine removal. However, this must be balanced against evidence suggesting that clinically indicated removal may increase infiltration and catheter occlusion rates. To ensure early detection of PVC complications and enable informed, timely removal decisions, regular assessments and documentation of PVCs at insertion sites and ongoing necessity checks at least once per shift are recommended. Clinical assessments should guide prompt PVC removal decisions in collaboration with discussions involving patients, their families and the multidisciplinary team.

Implications for research

Blinding of clinicians was not feasible given the visible nature of the intervention, as both groups followed distinct removal policies. However, none of the included studies reported systematic differences in infection prevention or clinical management between groups, so we judged the risk of performance bias to be low. In contrast, most outcomes, such as thrombophlebitis and infiltration, were assessed by clinical staff aware of group allocation, introducing potential detection bias due to the subjective nature of these assessments. We considered only microbiological outcomes (e.g. catheter-related bloodstream infection (CRBSI) and primary bloodstream infection (BSI)), which were assessed by blinded laboratory personnel, to be at low risk of detection bias. Future trials should capture better data on how easily routine removal is enacted. In addition, trials should measure clinical decision-making about removal, as prior studies have not asked clinicians why they do or do not comply with the routine or clinically indicated policy. As prior trials have tested the routine removal intervention up to 96 hours, future trials could examine routine removal at five days (120 hours) or longer. Ultrasound measurement is now commonly available in portable form at the bedside, and could be a potentially robust quantitative outcome measure for thrombophlebitis and infiltration in future trials, which may be less prone to bias than non-validated tools commonly used in previous trials.

Equity-related implications for research

Although exploratory subgroup analyses found no clinically meaningful differences between studies conducted in high-income and low- and middle-income countries for thrombophlebitis, infiltration or catheter occlusion, contextual factors such as staffing levels, healthcare infrastructure and infection prevention resources may still influence implementation. These findings suggest that, while the effect of clinically indicated removal appears similar across settings, further research in resource-limited contexts is needed to confirm equity-related implications and guide context-specific practice support.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD007798.pub6](https://doi.org/10.1002/14651858.CD007798.pub6).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Differences between the protocol and previous versions

Supplementary material 9 Routine removal versus clinically indicated removal of PVCs

Supplementary material 10 Clinically indicated versus routine removal of PVCs: thrombophlebitis - subgroup and sensitivity analyses

Supplementary material 11 Clinically indicated removal versus routine removal of PVCs: infiltration - subgroup and sensitivity analyses

Supplementary material 12 Clinically indicated removal versus routine removal of PVCs: catheter occlusion - subgroup and sensitivity analyses

ADDITIONAL INFORMATION

Acknowledgements

We acknowledge Joan Webster's contribution in the previous versions of the review. We acknowledge the contribution of The University of Queensland Information Specialist, Natalie Brown, who completed the systematic searches of the databases. We acknowledge Behzad Kiani for providing invaluable translation services.

Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Stavros K. Kakkos, University of Patras;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Anne-Marie Stephani, Cochrane Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Andrew Savage, Cochrane Editorial Service;
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Lindsay Robertson, Cochrane (methods review); Jo Platt, Central Editorial Information Specialist.

One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

Contributions of authors

KC: methodology, data curation, project administration, visualisation, writing – original draft

JS: methodology, data curation, visualisation, writing – original draft

NM: data curation, writing – review and editing

SR: data curation, visualisation, writing – original draft

SO: conceptualisation, data curation, writing – original draft

GM: data curation, formal analysis, visualisation, writing – original draft

SS: visualisation, writing – review and editing

CR: conceptualisation, methodology, data curation, writing – original draft

Declarations of interest

KC: none known

JS: Board membership of Intensive Care Foundation, Intensive Care Foundation Fiduciary Officer

NM: NM's employer has received unrestricted grants in aid from manufacturers of peripheral intravenous catheters and products on her behalf for academic research projects. These sponsors (Cardinal Health, 3M, Eloquest and Biolife) had no involvement in the study design, execution, analysis or publication of these projects and were unrelated to this review's topic. NM is an academic researcher and her employer has received funding on her behalf for her to provide expert advice or educational lectures on her research at professional symposia and other events from 3M, Medline and Wolter Kluwer. NM was also an investigator in some of the included trials. The assessment was allocated to review authors who were not investigators to eliminate potential conflicts of interest.

SO: none known

What's new

SR: none known

GM: none known

SS: none known

CMR: CMR's employer has received unrestricted research grants in aid from a manufacturer (BD) of peripheral intravenous catheters and products, on her behalf for academic research projects. This funder had no involvement in the study design, execution, analysis or publication of these projects, had no control over the projects, and they were unrelated to the topic of this review. CMR is an academic researcher and her employer has received funding on her behalf for her to provide consultancies (expert advice or educational lectures about her research at professional symposia) from manufacturers Solventum/3M, BBraun, BD and ITL Medical. CMR's employer has received unrestricted education grants from manufacturers (Solventum/3M, ICU Medical, Angiodynamics/Spectrum Vascular and ICU Medical) to support a research network that she established to fund travel grants to academic clinicians to present their own research (independent of funders). CR was also an investigator in some of the included trials. The assessment was allocated to review authors who were not investigators to eliminate potential conflicts of interest.

Sources of support

Internal sources

- This update of the review did not receive any funding, Other
Nil support received

External sources

- Chief Scientist Office, Scottish Government Health Directorates, The Scottish Government, UK

The Cochrane Vascular editorial base is supported by the Chief Scientist Office.

Registration and protocol

Protocol: 'Clinically indicated replacement versus routine replacement of peripheral venous catheters', DOI: 10.1002/14651858.CD007798 [31]

Data, code and other materials

The published Cochrane review provides the following resources for Cochrane Library users to download: comprehensive search strategies; citations for each unique report linked to all studies that were included, are ongoing, waiting for classification or excluded at the full-text screening in the final review; study data comprising information, study arms and results or test data; consensus risk of bias assessments; and analysis data that includes overall and subgroup estimates along with settings. Necessary permissions for this use have been secured. Analyses and data management were completed with Cochrane's authoring tool (RevMan), employing built-in computational methods. The authors can provide template data extraction forms from Excel upon reasonable request.

Date	Event	Description
27 January 2026	New search has been performed	New search run 21 January 2025; we included five new studies and excluded four new studies in this update.
27 January 2026	New citation required and conclusions have changed	Conclusions have been updated to reflect that clinically indicated removal of PVCs may increase infiltration and occlusion per participant, but likely results in little to no difference per 1000 days, compared to routine removal.

History

Protocol first published: Issue 2, 2009

Review first published: Issue 3, 2010

Date	Event	Description
23 August 2018	New search has been performed	Searches re-run. Two new studies included. No new studies excluded. One study awaiting classification.
23 August 2018	New citation required but conclusions have not changed	Searches re-run. Two new studies included. No new studies excluded. One study awaiting classification. Text updated. No change to conclusions.
25 March 2015	New citation required but conclusions have not changed	Searches re-run. No additional studies included or excluded. Minor changes to the text to adhere to current Cochrane standards.
25 March 2015	New search has been performed	Searches re-run. No additional studies included or excluded.

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ADDITIONAL TABLES

Table 1. Overview of included studies* and synthesis

Study with publication year (country)	Patient population	Point of randomisation	Control	Intervention	Critical outcomes	Important and limited importance outcomes
Barker, 2004 (United Kingdom)	Single site Adults Medical/surgical	Point of insertion	Clinically indicated removal	Routine removal - every 48 hours	—	1. Thrombophlebitis 2. Catheter per participant

Table 1. Overview of included studies* and synthesis (Continued)

Chin, 2018 (Australia)	Single site Neonates NICU	Time of scheduled replace- ment	Clinically in- dicated re- moval	Routine re- moval - every 72 to 96 hours	—	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Catheter per participant
Hossein- zadeh, 2014 (Iran)	Single site Paediatrics	At point of consent	Routine re- moval - every 72 hours	Clinically indicated removal	1. CRBSI 2. Local in- fection	1. Thrombophlebitis 2. Infiltration 3. Occlusion
Li, 2021 (China)	Multi-centre Adults Medical/surgi- cal	At point of consent	Routine re- moval - every 96 hours	Clinically indicated removal	1. CRBSI 2. Primary BSI 3. Local in- fection 4. Mortality	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Cost 5. Catheter per participant
Lin, 2021 (Taiwan)	Single site Paediatric Medical	Prior to consent	Routine re- moval - every 72 hours	Clinically indicated removal	—	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion
Lu, 2021 (China)	Single site Adults Medical/surgi- cal	At point of consent	Routine re- moval - every 96 hours	Clinically indicated removal	1. CRBSI 2. Local in- fection	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion
Nishanth, 2009 (India)	Single site Adults Surgical	At point of consent	Clinically in- dicated re- moval	Routine re- moval - every 48 hours	—	1. Thrombophlebitis 2. Catheter per participant
Rickard, 2010 (Australia)	Single site Adults Medical/surgi- cal	At point of consent	Routine re- moval - every 72 hours	Clinically indicated removal	1. CRBSI 2. Local in- fection	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Cost 5. Catheter per participant
Rickard, 2012 (Australia)	Multi-centre Adults Medical/surgi- cal	At point of consent	Routine re- moval - every 72 hours	Clinically indicated removal	1. CRBSI 2. Primary BSI 3. Local in- fection 4. Mortality	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Cost 5. Catheter per participant

Table 1. Overview of included studies* and synthesis *(Continued)*

Van Donk, 2009 (Australia)	Single centre Adults HITH	At point of consent	Routine removal - every 72 to 96 hours	Clinically indicated removal	1. CRBSI 2. Primary BSI 3. Local infection	1. Thrombophlebitis 2. Catheter occlusion
Ven-darmim, 2018 (Brazil)	Multi-centre Adults Medical/surgical	At point of consent	Routine removal - every 96 hours	Clinically indicated removal	1. CRBSI	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Pain 5. Satisfaction 6. Catheter per participant
Webster, 2007 (Australia)	Single site Adults Medical/surgical	At point of consent	Routine removal - every 72 hours	Clinically indicated removal	1. CRBSI 2. Local infection	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Cost 5. Pain
Webster, 2008 (Australia)	Single site Adults Medical/surgical	At point of consent	Routine removal - every 72 hours	Clinically indicated removal	1. CRBSI 2. Local infection	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion 4. Cost 5. Catheters per participant
Xu, 2017 (China)	Single site Adults Medical/surgical	Ward randomised (randomisation not at patient level)	Routine removal - every 72 hours	Clinically indicated removal	1. CRBSI 2. Local infection	1. Thrombophlebitis 2. Infiltration 3. Catheter occlusion

BSI: bloodstream infection; CRBSI: catheter-related bloodstream infection; HITH: hospital in the home; NICU: neonatal intensive care unit

*See [Supplementary material 2](#) for full details of the included studies.